

Multivariate Witness-Hiding Adaptor Signatures

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Abstract. Adaptor signatures extend digital signatures with conditional disclosure capabilities, enabling atomic swaps, payment channels, and other advanced blockchain protocols. Although post-quantum adaptor signatures have been explored under lattice, isogeny, and coding-theoretic assumptions, no constructions have yet been realised from the multivariate quadratic (MQ) family of signatures. Classical algebraic adaptor techniques rely on embedding the witness into signing randomness, which is natural for discrete-log-based schemes but does not apply to MQ signatures such as UOV and MAYO: to the best of our knowledge, MQ signing randomness lacks the algebraic structure needed for witness embedding, and no such algebraic adaptor construction is currently known. This motivates a different approach. We propose MWAS, the first commitment-based adaptor-style construction for MQ signatures, specifically UOV and MAYO from the NIST PQC process, implemented via the Open Quantum Safe library. Our construction uses a lightweight HMAC-SHA256 commitment and a concatenation-based adaptation, supporting a hash-preimage witness relation. We prove correctness, witness hiding, and witness extractability in the ROM under MQ-hardness and PRF assumptions. A prototype implementation on a Raspberry Pi 5 shows pre-signature generation under 0.4 ms for UOV and 0.5–3.3 ms for MAYO across 128–256-bit security levels, with throughput up to 710 ops/s and public key sizes of 1.4 KB (MAYO) to 1.2 MB (UOV). These results indicate that commitment-based MQ adaptor signatures are a viable post-quantum option for settings where hash-preimage witness relations are appropriate.

Keywords: Post-Quantum Cryptography · Adaptor Signatures · Multivariate Cryptography · Open Quantum Safe · Witness Hiding

1 Introduction

The shift to the post-quantum age has become an immediate engineering challenge rather than a mere theoretical issue. Despite the current lack of cryptographically significant quantum computers, the risk of harvest-now, decrypt-later attacks has generated an urgent need to implement quantum-resistant primitives. Critical infrastructure, financial systems, and blockchain protocols currently face long-lasting threats if encrypted data obtained today can be decoded in the future. This urgency has expedited worldwide standardisation initiatives, particularly the NIST Post-Quantum Cryptography (PQC) process [AASA⁺20], which has propelled lattice-based schemes such as Kyber for key encapsulation and Dilithium for digital signatures toward extensive adoption. A substantial dependence on a single set of hardness assumptions generates systemic risk. To ensure durability, the post-quantum cryptographic ecosystem must incorporate diversity in its mathematical underpinnings, including multivariate, hash-based, and code-based primitives.

However, transitioning to post-quantum security requires more than simply replacing traditional primitives. Modern applications increasingly demand advanced cryptographic functionalities such as adaptor signatures [PBF⁺18, EEE20], which enable conditional

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disclosure mechanisms essential for atomic swaps, payment channels, and scriptless smart contracts in blockchain ecosystems. These protocols rely on signatures that can be partially completed if a hidden witness is revealed, while guaranteeing the deterministic extraction of the witness from the completed signature [FS90]. Preserving these properties in the post-quantum era is critical to ensuring the security and functionality of next-generation distributed systems.

Despite the importance of post-quantum adaptor signatures, existing research has focused primarily on lattice-based instantiations, particularly those built on Dilithium and related NIST-standardised signatures [EEE20, TMSM21b, QCY21]. Some work has been extended to isogeny-based approaches [TMSM21b] and coding-theoretic approaches [KH22], but the MQ family of signatures [Pat97, Beu21] has been largely overlooked. A key reason is that classical algebraic adaptor techniques — where the witness is embedded into signing randomness — do not naturally carry over to MQ schemes: to the best of our knowledge, no algebraic adaptor construction for UOV or MAYO is currently known. This motivates a commitment-based approach, where the witness is committed to via HMAC-SHA256 and incorporated into the signed message. Overreliance on lattice-based solutions not only risks cryptographic monoculture but also imposes practical limitations: lattice-based adaptor signatures often have prohibitive bandwidth requirements for resource-constrained or high-throughput environments, and few production-grade, constant-time implementations with integration into libraries such as `liboqs` currently exist. In contrast, MQ schemes such as UOV and MAYO [Hei25] offer complementary hardness assumptions, compact signature sizes, and efficient verification, making them natural candidates for commitment-based adaptor-style constructions.

This work addresses the existing gap by proposing MWAS (Multivariate Witness-Hiding Adaptor Signatures), a commitment-based adaptor-style construction for MQ signatures, specifically UOV [Pat97] and MAYO [Beu21]. Using a commit-then-sign design, MWAS achieves the core adaptor signature properties — correctness, extractability, and witness hiding — through the binding and pseudorandom properties of HMAC-SHA256, without requiring algebraic structure in the base scheme. Our contributions are as follows.

- **First MQ-based adaptor signature construction:** To the best of our knowledge, we present the first adaptor signature scheme for MQ-based signatures. We formalise MWAS and prove correctness, aEUF-CMA security, witness hiding, and witness extractability in the ROM.
- **Formal security analysis:** Security is reduced to MQ-hardness, the PRF security of HMAC-SHA256, and collision resistance of SHA-256, extending the adaptor signature framework to a non-algebraic, non-lattice setting.
- **Constant-time C implementation:** We implement MWAS using `liboqs` for UOV and MAYO across 128-, 192-, and 256-bit security levels, with constant-time operations and secure memory handling.
- **Performance evaluation:** We benchmark end-to-end timings, throughput, and signature sizes on a Raspberry Pi 5 and compare results against prior post-quantum adaptor signature schemes across all security levels.

The remainder of this paper is structured as follows. Section 2 reviews related work and identifies gaps in existing adaptor signature constructions. Section 3 introduces preliminaries, including notation, basic primitives, the MQ problem, and detailed descriptions of UOV and MAYO. Section 4 presents our construction. Section 5 formalises the security model, definitions, and games, and provides lemmas, theorems, and proof sketches. Section 6 reports parameter analysis, implementation details, and experimental results, including performance under varying network latencies. Section 7 discusses broader implications, and Section 8 concludes.

2 Related Work and Gap Analysis

The notion of adaptor signatures originated in blockchain research, with [Fou19] introducing “one-time verifiably encrypted signatures” to enable trustless fair exchange in cryptocurrency protocols. [AEE⁺21] generalised this idea, formalising adaptor signatures with comprehensive security definitions and demonstrating their applicability to Bitcoin-compatible payment channels and multiparty protocols [EFH⁺21]. In addition, adaptor signatures can be systematically derived from identifying protocols through the Fiat–Shamir transformation, establishing them as a general cryptographic primitive.

Subsequent work addressed the limitations in the original security models. [DOY22] identified that the definitions in [AEE⁺21] were insufficient for multiquery scenarios, introducing stronger notions such as extractability, unique extractability, and unlinkability. Later, [GSST24] highlighted concrete vulnerabilities, including malleable and leaky pre-signatures, and introduced pre-verify soundness as a necessary property. Most recently, [LTZ24] refined the framework by identifying witness exposure in adapted signatures and proposing witness-hiding constructions using trapdoor commitments, supporting arbitrary NP relations. MWAS is closest in spirit to [LTZ24], but differs in three key respects: (i) [LTZ24] is scheme-agnostic and supports any NP relation via trapdoor commitments, whereas MWAS is a concrete instantiation for the hash-preimage relation over HMAC-SHA256; (ii) MWAS uses a lightweight HMAC commitment rather than a trapdoor commitment, trading generality for simplicity and efficiency; and (iii) extraction in MWAS is a direct parse of the concatenated witness, relying on the binding property of HMAC rather than the algebraic structure of a trapdoor. MWAS thus complements [LTZ24] by providing the first MQ-based instantiation. The advent of quantum computing motivated the study of post-quantum adaptor signatures across various cryptographic assumptions. [EEE20] pioneered lattice-based constructions from Module-SIS and Module-LWE, though their LAS scheme suffered from weak adaptability and privacy leakage where witness norms revealed path positions. [TMSM21b] advanced the state-of-the-art with isogeny-based constructions (CSI-FiSh), achieving full adaptability, privacy preservation, and a three-fold reduction in chain storage compared to lattice schemes. Their IAS scheme employed uniformly distributed witnesses to prevent leakage in payment applications. [KH22] contributed coding-theoretic variants via McEliece-based signatures, offering the smallest pre-signatures among post-quantum approaches. More recently, [RK24] introduced SQIAsignHD, which leverages supersingular isogenies and utilises insights from SIDH attacks to extract witnesses while mitigating vulnerabilities through orientation techniques.

Practical applications have underscored the transformative role of adaptor signatures in blockchain scalability and interoperability. [TMSM21a] proposed A²L (Anonymous Atomic Locks), the first payment-channel hub protocol combining security, privacy, interoperability, and fungibility, with a thirty-three-fold bandwidth reduction over TumbleBit. [QPM⁺23] introduced Blind Adaptor Signatures to overcome Bitcoin-compatible hub limitations, enabling variable amounts while preserving anonymity and value privacy. Atomic swaps received particular attention: [TMMS22] presented universal atomic swaps based on verifiable timed discrete logarithms, and [HdP21] achieved the first Bitcoin–Monero deployment, solving a long-standing challenge for trustless exchange between transparent and privacy-focused cryptocurrencies. Optimisations for batched swaps were presented in [TZY22], reducing the ECDSA-based adaptor signature overhead through efficient zero-knowledge batching. In parallel, [SLYQ22] introduced MONET, enabling scriptless Monero payment channels.

Recent advances extend adaptor signatures to broader functionality. [VST24] proposed functional adaptor signatures (FAS), enabling fair functional sales where buyers learn only function outputs without accessing inputs, facilitating privacy-preserving machine learning and statistical applications. [RL24] developed Bitcoin Clique with two-shot adaptor signatures, which enforce honest behaviour by exposing keys if conflicting signatures are

published. [CLTZ25] achieved universal constructions using MPC-in-the-Head, avoiding costly reductions while maintaining black-box security for arbitrary NP relations. Finally, [TM21] generalised adaptor signatures into lockable signatures, supporting arbitrary schemes through scriptless scripts. The supporting infrastructure and analytical frameworks have evolved in parallel with core constructions. [TMSS23] introduced *LedgerLocks*, a comprehensive security framework for blockchain protocols built on adaptor signatures, providing formal verification tools for complex multiparty settings. Key management issues were addressed in [ER22] through the use of deterministic wallets for adaptor signatures. Surveys such as [BDE⁺23] placed adaptor signatures within the broader family of exotic post-quantum signatures, while [NK24] focused specifically on quantum-proof blockchain security. Extensions to multi-party settings include threshold constructions [JXGZ24], identity-based compatibility [ZHB⁺23], and randomised integration of EdDSA [ZLLY25]. The interoperability foundations of cross-chains were studied in [ABC⁺24] and [ZABZ⁺21]. Recent theoretical advances include DSKE (Digital Signatures with Key Extraction) [WAK⁺25] and multi-round proof frameworks with adaptor signature applications [ABB⁺24]. Novel application domains are also emerging: [FS25] applied adaptor signatures to high-frequency trading for nearly instantaneous cross-chain trades, [YKN25] proposed hybrid stabilisation protocols leveraging AI-driven arbitrage, and [HPDB25] introduced a formal fairness framework for signature exchange.

The evolution of adaptor signatures from informal blockchain scaling mechanisms to mature cryptographic primitives demonstrates the convergence of theoretical foundations and real-world deployment. Progress has followed a clear trajectory: foundational formalisation, refinement of security models, and large-scale practical applications. Current research frontiers focus on universal applicability, post-quantum efficiency, and eventual standardisation: positioning adaptor signatures as fundamental components of next-generation blockchain infrastructures, enabling scalable, interoperable, and privacy-preserving systems.

2.1 Existing Adaptor Signature Constructions

Post-quantum adaptor signatures have primarily been investigated in the lattice setting. Esgin et al. [EEE20] pioneered lattice-based constructions for payment channel protocols, proving feasibility but with substantial overhead in artefact size and verification cost. Subsequent work, such as [TMSM21b], refined these ideas for off-chain, privacy-preserving payments; however, bandwidth demands remain high in both constrained and high-throughput environments.

Orthogonal directions include generic frameworks that decouple the adaptor layer from the base signature [QCY21], coding-theoretic instantiations [KH22], threshold and multiparty variants [JXGZ24], and functional adaptor signatures that enable selective disclosure [VST24]. Despite this breadth, little work has explored constructions from MQ assumptions, leaving schemes such as UOV [KPG99] and MAYO [Beu21] promising but largely untapped candidates for adaptor signature frameworks.

2.1.1 Identified Research Gaps

1. **Lack of MQ-based constructions:** To the best of our knowledge, no adaptor signature construction — algebraic or otherwise — has been proposed for multivariate quadratic (MQ) schemes such as UOV or MAYO. Classical algebraic embedding of witnesses into signing randomness is not known to apply to these schemes, and no commitment-based alternative has been formalised [KPG99].
2. **Witness hiding in non-lattice settings:** Prior work has focused primarily on extractability; strong witness-hiding guarantees outside of lattice-based frameworks remain underexplored.

3. **Implementation maturity:** Existing implementations rarely achieve constant-time security with robust memory handling. Integration into established PQC libraries (e.g., `liboqs`) is still lacking.
4. **Advanced properties over new foundations:** While threshold and functional extensions have been considered in lattice-based schemes, their applicability to MQ-based designs has not yet been investigated.

MWAS directly addresses each of these gaps; see Section 4 for the construction and Section 5 for the formal security analysis.

3 Preliminaries

This section establishes the mathematical foundations and cryptographic primitives underlying our multivariate witness-hiding adaptor signature construction. We present the core definitions, security models, hardness assumptions, and detailed descriptions of the multivariate schemes essential to our construction.

3.1 Notation and Basic Cryptographic Primitives

In this study, we represent the security parameter as $\lambda \in \mathbb{N}$, which establishes the computational security level of our schemes. A function $\text{negl}(\lambda)$ is considered negligible if, for any positive constant c , there exists a threshold $\lambda_0 \in \mathbb{N}$ such that for all $\lambda > \lambda_0$, it holds that $\text{negl}(\lambda) < \lambda^{-c}$. A probabilistic polynomial-time (PPT) algorithm is a randomised algorithm with a running time bounded by a polynomial in λ . A quantum polynomial-time (QPT) algorithm refers to its quantum counterpart, utilising quantum computational resources [Gol01]. The message space is represented by $\mathcal{M}$.

We denote the finite field with q elements, where q is a prime power, by $\mathbb{F}_q$. Notation $x \xleftarrow{\$} S$ indicates that x is sampled uniformly at random from a set S , while $x \leftarrow \mathcal{A}(y)$ denotes the output of algorithm $\mathcal{A}$ on input y . We write $\{0, 1\}^n$ for the set of all binary strings of length n , and $\{0, 1\}^*$ for the set of all finite binary strings. The concatenation of two strings a and b is written as $a \parallel b$. Vectors are denoted in bold lowercase, e.g., $\mathbf{v} \in \mathbb{F}_q^n$, and matrices in bold uppercase, e.g., $\mathbf{M} \in \mathbb{F}_q^{m \times n}$.

A digital signature scheme $\Sigma = (\text{KeyGen}, \text{Sign}, \text{Verify})$ comprises the following PPT algorithms. The key generation algorithm $\text{KeyGen}(1^\lambda)$ accepts the security parameter in unary format and generates a key pair (sk, pk) , where sk is the secret signing key and pk is the public verification key. The signing algorithm $\text{Sign}(sk, m)$ accepts a secret key and a message $m \in \mathcal{M}$, generating a digital signature σ . The deterministic verification algorithm $\text{Verify}(pk, m, \sigma)$ yields 1 if σ constitutes a valid signature for m with respect to pk , and 0 otherwise. The objective of a digital signature scheme is to ensure authenticity and integrity: only the holder of sk can generate valid signatures, while anyone with pk can verify them.

The correctness property stipulates that for every message $m \in \mathcal{M}$ and all legitimately created key pairs $(sk, pk) \leftarrow \text{KeyGen}(1^\lambda)$, the following hold:

$$\Pr[\text{Verify}(pk, m, \text{Sign}(sk, m)) = 1] = 1$$

The standard security notion for digital signatures is the existential unforgeability under chosen-message attacks (EUF-CMA) [GMR88]. A signature scheme Σ is EUF-CMA secure if for all PPT adversaries $\mathcal{A}$ with access to a signing oracle $\mathcal{O}_{\text{Sign}}$, the probability that $\mathcal{A}$ produces a valid signature on a message not previously enquired from the oracle is negligible:

$$\text{Adv}_{\Sigma, \mathcal{A}}^{\text{EUF-CMA}}(\lambda) = \Pr[\mathcal{A}^{\mathcal{O}_{\text{Sign}}}(pk) \text{ outputs valid forgery}] \leq \text{negl}(\lambda)$$

A *commitment scheme* enables a party to pledge a value while keeping it secret [Nao91], with the possibility of revealing it later in a way that can be verified. Formally, two PPT algorithms ($\text{Commit}, \text{Verify}$) make up a commitment scheme $\mathcal{C}$:

- $\text{Commit}(m, r) \rightarrow (c, d)$: Given a message m and randomness r , the algorithm outputs a commitment c and the corresponding decommitment information d .
- $\text{Verify}(c, m, d) \rightarrow \{0, 1\}$: Given a commitment c , message m , and decommitment d , the algorithm outputs 1 if d correctly opens c to m , and 0 otherwise.

Correctness requires that for all m, r , if $(c, d) \leftarrow \text{Commit}(m, r)$, then $\text{Verify}(c, m, d) = 1$.

A secure commitment scheme must satisfy two fundamental properties. The property *binding* ensures that no PPT adversary can open a commitment to two distinct messages:

$$\Pr \left[\mathcal{A}(1^\lambda) \rightarrow (c, m_0, d_0, m_1, d_1) : m_0 \neq m_1 \wedge \text{Verify}(c, m_0, d_0) = 1 \wedge \text{Verify}(c, m_1, d_1) = 1 \right] \leq \text{negl}(\lambda)$$

The hiding property ensures that no information about the committed value is disclosed. For computational hiding, for all PPT distinguishers $\mathcal{D}$ and all messages m_0, m_1 :

$$|\Pr[\mathcal{D}(\text{Commit}(m_0, r_0)) = 1] - \Pr[\mathcal{D}(\text{Commit}(m_1, r_1)) = 1]| \leq \text{negl}(\lambda)$$

The Random Oracle Model [BR93], which is used in our security analysis, models cryptographic hash functions as genuinely random functions accessible only through oracle queries. By acting as a random mapping, the hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ in the ROM produces outputs that are uniformly distributed and unaffected by different inputs. This idealisation makes it possible for rigorous security proofs while capturing the essential unpredictability of well-designed hash functions.

The HMAC construction [BCK96b, BCK96a], which plays a central role in our commitment scheme, combines a hash function that uses a secret key to offer pseudo-randomness and authentication. HMAC is defined as follows, given a hash function H and key k :

$$\text{HMAC}_k(m) = H((k \oplus \text{opad}) \| H((k \oplus \text{ipad}) \| m))$$

where $\text{opad} = 0x5C$ and $\text{ipad} = 0x36$ are fixed padding constants repeated to the block length. Assuming that H operates as a random oracle, HMAC provides both pseudorandom function (PRF) security and strong unforgeability. The double application of the hash function in HMAC provides resistance against length extension attacks.

3.2 Adaptor Signature Schemes

By embedding a hard relation $R \subseteq \mathcal{W} \times \mathcal{Y}$, where $\mathcal{W}$ is the witness space and $\mathcal{Y}$ is the statement space, an adaptor signature scheme extends a conventional digital signature scheme. The NP language is induced by the relation R :

$$L_R = \{Y \in \mathcal{Y} \mid \exists w \in \mathcal{W} : (w, Y) \in R\}$$

These PPT algorithms make up an adaptor signature scheme:

- $\text{KeyGen}(1^\lambda) \rightarrow (sk, pk)$: Generates signing and verification keys.
- $\text{PreSign}(sk, m, Y) \rightarrow \hat{\sigma}$: Produces a pre-signature $\hat{\sigma}$ for message m , which embeds the statement Y but hides its witness.
- $\text{PreVerify}(pk, m, Y, \hat{\sigma}) \rightarrow \{0, 1\}$: Checks whether $\hat{\sigma}$ is a valid pre-signature for (m, Y) .

- $\text{Adapt}(Y, w, \hat{\sigma}) \rightarrow \sigma$: $\hat{\sigma}$ is transformed into a complete signature σ given a valid witness w with $(w, Y) \in R$.
- $\text{Extract}(Y, \sigma, \hat{\sigma}) \rightarrow w$: From a valid pair $(\sigma, \hat{\sigma})$, the embedded witness w is extracted.

Every legitimate pre-signature $\hat{\sigma}$ and witness w with $(w, Y) \in R$ must meet the following requirements to be considered correct:

$$\text{Verify}(pk, m, \sigma) = 1, \text{ where } \sigma = \text{Adapt}(Y, w, \hat{\sigma})$$

Assume that the adaptor signature scheme $\mathcal{AS} = (\text{KeyGen}, \text{PreSign}, \text{PreVerify}, \text{Adapt}, \text{Verify}, \text{Extract})$ is based on the secure signature scheme Σ . The primary security attributes are as follows:

Pre-signature Correctness requires:

$$\text{PreVerify}(pk, m, Y, \text{PreSign}(sk, m, Y)) = 1$$

Adaptability requires:

$$\text{Verify}(pk, m, \text{Adapt}(Y, w, \hat{\sigma})) = 1, \text{ for all } (w, Y) \in R$$

Witness Extractability. There must exist an efficient extractor such that:

$$\text{Extract}(Y, \sigma, \hat{\sigma}) = w$$

for every tuple $(Y, \sigma, \hat{\sigma})$ satisfying:

$$\hat{\sigma} = \text{PreSign}(sk, m, Y), \quad \sigma = \text{Adapt}(Y, w, \hat{\sigma}).$$

Witness Hiding. Witness hiding guarantees that no (PPT) adversary $\mathcal{A}$ can recover a valid witness w' from a simulated pre-signature:

$$\Pr[w' \leftarrow \mathcal{A}(pk, m, Y, \hat{\sigma}) : (w', Y) \in R] \leq \text{negl}(\lambda).$$

aEUF-CMA Security. Let $\mathcal{A}$ be a PPT adversary with oracle access to both PreSign and Adapt . After making a polynomial number of queries, $\mathcal{A}$ produces a tuple (m^*, Y^*, σ^*) and is said to win if the following conditions are met.

1. $\text{Verify}(pk, m^*, \sigma^*) = 1$,
2. (m^*, Y^*) was never queried with a known witness, and
3. $\mathcal{A}$ does not know any w^* such that $(w^*, Y^*) \in R$.

The scheme is *aEUF-CMA secure* if:

$$\Pr[\mathcal{A} \text{ wins}] \leq \text{negl}(\lambda).$$

An adaptor signature scheme is considered secure if it satisfies all of the aforementioned properties: pre-signature correctness, adaptability, witness extractability, witness hiding, and aEUF-CMA security.

3.3 The MQ Problem and Cryptographic Assumptions

The computational intractability of solving MQ systems over finite fields forms the basis of security for several post-quantum signature schemes, such as UOV and MAYO [Hei25, DGS06]. Formally, let $\mathbb{F}_q$ be a finite field, and let $m, n \in \mathbb{N}$. An instance of the MQ problem comprises m quadratic equations in n variables with coefficients in $\mathbb{F}_q$.

Consider a system of quadratic polynomials.

$$P_i(x_1, \dots, x_n) \in \mathbb{F}_q[x_1, \dots, x_n], \quad i = 1, \dots, m,$$

where each polynomial P_i has degree at most 2, and a target vector $\mathbf{y} = (y_1, \dots, y_m) \in \mathbb{F}_q^m$. The goal is to find a vector

$$\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_q^n$$

such that

$$\mathbf{P}(\mathbf{x}) = (P_1(\mathbf{x}), \dots, P_m(\mathbf{x})) = \mathbf{y}.$$

Each polynomial P_i can be expressed in the following general form:

$$P_i(x_1, \dots, x_n) = \sum_{1 \leq j \leq k \leq n} \alpha_{ijk} x_j x_k + \sum_{j=1}^n \beta_{ij} x_j + \gamma_i,$$

where $\alpha_{ijk}, \beta_{ij}, \gamma_i \in \mathbb{F}_q$ are coefficients.

For appropriate choices of parameters (m, n, q) where typically $m \geq n$ and the system is overdetermined or random, the MQ problem is computationally intractable for any PPT adversary. The MQ problem is known to be NP-hard in the worst case, and no algorithm is currently known that solves random MQ instances in polynomial time with non-negligible probability. Formally, given a system of uniformly random m quadratic polynomials in n variables over $\mathbb{F}_q$, the probability that a PPT adversary $\mathcal{A}$ finds a valid solution is negligible in λ :

$$\Pr[\mathcal{A}(P_1, \dots, P_m, \mathbf{y}) = \mathbf{x} \text{ s.t. } \mathbf{P}(\mathbf{x}) = \mathbf{y}] \leq \text{negl}(\lambda)$$

The security of most multivariate public-key methods is predicated on this premise, where the signer solves a structured MQ instance using trapdoor knowledge (namely, the secret key). In contrast, inversion without this trapdoor is assumed to be hard. The structured systems used in schemes like UOV and MAYO maintain the appearance of randomness to external observers while containing hidden algebraic structures that enable efficient solution with the secret key.

The security of our construction rests on two fundamental assumptions. First, we assume the EUF-CMA security of the underlying multivariate signature scheme ($\Sigma \in \{\text{UOV}, \text{MAYO}\}$) in the ROM. Under this assumption, given a signature scheme Σ based on the MQ problem, no PPT adversary can produce a valid forgery with access to a signing oracle:

$$\text{Adv}_{\Sigma, \mathcal{A}}^{\text{MQ-EUF-CMA}}(\lambda) = \Pr[\mathcal{A}^{\mathcal{O}_{\text{Sign}}}(pk) \rightarrow (m^*, \sigma^*) : \text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin \mathcal{Q}] \leq \text{negl}(\lambda).$$

where $\mathcal{Q}$ denotes the set of messages queried to the signing oracle.

Second, we assume the PRF security of HMAC-SHA256 [BCK96b, BCK96a], which ensures that HMAC outputs are computationally indistinguishable from truly random strings. For all PPT distinguishers $\mathcal{D}$ with oracle access:

$$\left| \Pr[\mathcal{D}^{\text{HMAC}_k(\cdot)}(1^\lambda) = 1] - \Pr[\mathcal{D}^{f(\cdot)}(1^\lambda) = 1] \right| \leq \text{negl}(\lambda),$$

where $f : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ is a truly random function.

3.4 Multivariate Signature Schemes: UOV and MAYO

3.4.1 The UOV Scheme

After Patarin's initial Oil and Vinegar scheme [Pat97] was cryptanalysed, Kipnis, Patarin, and Goubin [KPG99] developed the Unbalanced Oil and Vinegar Signature Scheme (UOV). The variant *unbalanced* with $v > o$ has defied cryptanalysis for more than 20 years and is one of the most reliable MQ designs; the balanced variant ($v = o$) was subjected to the Kipnis and Shamir [KS98] linearisation attack.

Mathematical Foundation. With parameters (v, o, q) , UOV is defined over a finite field $\mathbb{F}_q$: $m = o$ equations, $n = v + o$ total variables, v vinegar variables, and o oil variables.

Core Algebraic Structure. A set of m quadratic polynomials defines the core map $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$.

$$f_i(x) = \sum_{j,k \leq v} \alpha_{ijk} x_j x_k + \sum_{j \leq v < k \leq n} \beta_{ijk} x_j x_k + \sum_{j=1}^n \gamma_{ij} x_j + \delta_i.$$

Crucially, there are *no* oil–oil quadratic terms ($x_j x_k$ with $j, k > v$). In matrix form,

$$\mathbf{F}_i = \begin{pmatrix} \mathbf{A}_i & \mathbf{B}_i \\ \mathbf{B}_i^\top & \mathbf{0} \end{pmatrix},$$

with $\mathbf{A}_i \in \mathbb{F}_q^{v \times v}$ and $\mathbf{B}_i \in \mathbb{F}_q^{v \times o}$. The quadratic part of f_i can be written as $x^\top \mathbf{F}_i x$; together with linear and constant terms, $f_i(x) = x^\top \mathbf{F}_i x + \gamma_i^\top x + \delta_i$.

Oil Space and Trapdoor Intuition. Let $\mathcal{O}_F = \text{span}\{e_{v+1}, \dots, e_n\} \subseteq \mathbb{F}_q^n$ denote the (standard) oil subspace. For inputs supported on $\mathcal{O}_F$, the *quadratic* contribution vanishes (no oil–oil block), but *linear* oil terms generally remain; thus, $F(\mathbf{o})$ does not equal $\mathbf{0}$ in general. The trapdoor comes from the fact that, after fixing the vinegar variables, each f_i becomes *linear* in the oil variables.

Affine Hiding Transformation. Select an affine map that can be inverted to conceal the structure $T : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$, $T(x) = \mathbf{T}_{\text{lin}} x + \mathbf{t}$ with $\mathbf{T}_{\text{lin}} \in \text{GL}_n(\mathbb{F}_q)$. $P = F \circ T$ is the public key. Write each polynomial in quadratic form as

$$f_i(y) = y^\top A_i y + b_i^\top y + c_i, \quad \text{with } y = T(x),$$

The corresponding public equations can be expressed as

$$p_i(x) = f_i(T(x)) = x^\top P_i x + L_i^\top x + c'_i,$$

where

$$P_i = \mathbf{T}_{\text{lin}}^\top A_i \mathbf{T}_{\text{lin}}, \quad L_i^\top = (2 \mathbf{T}_{\text{lin}}^\top A_i \mathbf{t} + \mathbf{T}_{\text{lin}}^\top b_i)^\top,$$

and

$$c'_i = \mathbf{t}^\top A_i \mathbf{t} + b_i^\top \mathbf{t} + c_i.$$

Remark. In characteristic 2, the factor 2 disappears.

3.4.2 Key Generation

Algorithm 1 UOV Key Generation (private and public)

- 1: Sample a central map $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ with no oil–oil quadratic terms
 - 2: Sample $\mathbf{T}_{\text{lin}} \xleftarrow{\$} \text{GL}_n(\mathbb{F}_q)$ and $\mathbf{t} \xleftarrow{\$} \mathbb{F}_q^n$
 - 3: Define $T(x) = \mathbf{T}_{\text{lin}}x + \mathbf{t}$ and $P = F \circ T$
 - 4: **Output:** secret key (F, T) ; public key $P = \{p_i\}_{i=1}^m$
-

3.4.3 Signing Procedure

Algorithm 2 Procedure: UOV Signing

- 1: **Inputs:** a message M and the private key (F, T)
 - 2: Compute target hash $\mathbf{h} \leftarrow H(M) \in \mathbb{F}_q^m$
 - 3: **repeat**
 - 4: Sample vinegar vector $\mathbf{v} \xleftarrow{\$} \mathbb{F}_q^v$
 - 5: Substitute $\mathbf{v}$ into F ; derive a linear system in oil variables $\mathbf{o} \in \mathbb{F}_q^o$
 - 6: Attempt to solve $\mathbf{M}_{\text{oil}}\mathbf{o} = \mathbf{h} - F_{\text{vinegar}}(\mathbf{v})$
 - 7: **until** system is solvable
 - 8: Form $\mathbf{y} \leftarrow (\mathbf{v}, \mathbf{o})$; compute $\mathbf{s} \leftarrow T^{-1}(\mathbf{y})$
 - 9: **Output:** resulting signature $\mathbf{s} \in \mathbb{F}_q^n$
-

Notation for Algorithm 2. $\mathbf{M}_{\text{oil}}$ denotes the $o \times o$ matrix of the linear system in oil variables obtained after substituting the vinegar vector $\mathbf{v}$, and $F_{\text{vinegar}}(\mathbf{v})$ is the vinegar-only contribution to the right-hand side.

Linear System Analysis. Once $\mathbf{v}$ is fixed, each $f_i(\mathbf{v}, \mathbf{o})$ becomes linear in the oil variables, resulting in an $o \times o$ system ($m = o$). On average, fewer than two attempts are needed, as this is a full-rank system with a chance of approximately $1/q$ [Hei25].

3.4.4 Verification Procedure

Algorithm 3 Procedure: UOV Verification

- 1: **Inputs:** a message M , a candidate signature $\mathbf{s} \in \mathbb{F}_q^n$, and the public key $P = \{p_i\}_{i=1}^m$
 - 2: First, check that $\mathbf{s}$ is a valid element of $\mathbb{F}_q^n$; if not, **return Reject**
 - 3: Compute challenge hash $\mathbf{h} \leftarrow H(M) \in \mathbb{F}_q^m$
 - 4: **for** $i = 1$ to m **do**
 - 5: Evaluate $h'_i \leftarrow p_i(\mathbf{s})$
 - 6: **end for**
 - 7: **Decision:** Accept if $\mathbf{h}' = \mathbf{h}$, otherwise Reject
-

Complexity. Verification costs $O(mn^2)$ field operations (evaluating m quadratics). Signing is dominated by solving an $o \times o$ linear system (expected constant number of trials). Key generation costs $O(n^3)$.

3.4.5 Security Summary

MQ hardness. Given random looking P , solving $P(x) = h$ is MQ hard. **Indistinguishability.** $P = F \circ T$ should be computationally indistinguishable from random quadratics

(without visible oil-vinegar pattern). **Attacks.** Linearisation (balanced OV) is mitigated by $v > o$; MinRank / differential / Gröbner attacks have exponential costs in $(v - o)$ for standard parameters. **Quantum.** Grover gives a quadratic speedup; parameters choose $(v - o)$ large enough to retain NIST-level security.

3.4.6 Parameter Selection (Practitioner Notes)

- Security grows with $(v - o)$; signatures are $n = v + o$ field elements; PK size $\Theta(mn^2)$.
- Smaller q speeds up arithmetic, but typically requires larger $(v - o)$.
- Use vetted parameter sets and cite their sources; ensure that the numbers match the benchmarks reported in Section 6.

3.4.7 The MAYO Scheme

The MAYO scheme, introduced by Beullens [Beu21], refines UOV to achieve dramatically smaller public keys while retaining efficient signing and verification. Its central idea is to *shrink* the oil dimension o (far below the number of equations m) and then *restore solvability* via a public *whipping* transformation that mixes multiple copies of the variable space. This preserves the "no oil-oil quadratic" structure while enabling $o \ll m$ without sacrificing signing feasibility.

Mathematical Foundation. The tuple of parameters (n, m, o, k, q) specifies a MAYO scheme, where the system is defined over a finite field $\mathbb{F}_q$ with $n = v + o$ total variables. The design includes m quadratic equations, a deliberately small number of oil variables o , and an integer whipping factor $k \geq 1$. In contrast to UOV, where typically $m = o$, the MAYO approach employs the whipping mechanism to expand the number of effective oil variables to ko , ensuring solvability whenever $ko \geq m$.

Core Maps. As in UOV, choose a central quadratic map $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ with *no oil-oil quadratic terms*, and an invertible affine transformation $T(x) = \mathbf{T}_{\text{lin}}x + \mathbf{t}$. The public base map is $P = F \circ T$. For vectors supported in the oil subspace, quadratic contributions vanish but linear oil terms remain, so $P(\mathbf{o})$ is not identically zero.

Whipping Transformation. In MAYO, the signing map is extended by lifting the variable space into k parallel blocks. The public map that is produced is

$$P^* : (\mathbb{F}_q^n)^k \longrightarrow \mathbb{F}_q^m, \quad P^*(x_1, \dots, x_k) = \sum_{i=1}^k E_{ii} P(x_i) + \sum_{1 \leq i < j \leq k} E_{ij} \Delta_P(x_i, x_j),$$

where the bilinear form $\Delta_P(x, y) := P(x + y) - P(x) - P(y)$ arises as the polarisation of P , and the matrices $E_{ij} \in \mathbb{F}_q^{m \times m}$ (called *emulsifiers*) are fixed and publicly known. Using an appropriate choice of $\{E_{ij}\}$, the dimension of the oil space is effectively expanded to ko , ensuring solvability whenever $ko \geq m$.

Emulsifier Design. The emulsifiers (i) expand the dimension to ensure solvability ($ko \geq m$), (ii) prevent algebraic leakage, and (iii) maintain compact representation (system-wide, not per key). The constructions follow [Beu21, Hei25].

3.4.8 Key Generation

Algorithm 4 MAYO Key Generation

- 1: Sample $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ with no oil–oil quadratic terms
 - 2: Sample $\mathbf{T}_{\text{lin}} \xleftarrow{\$} \text{GL}_n(\mathbb{F}_q)$ and $\mathbf{t} \xleftarrow{\$} \mathbb{F}_q^n$; set $T(x) = \mathbf{T}_{\text{lin}}x + \mathbf{t}$
 - 3: Set $P = F \circ T$; fix public emulsifiers $\{E_{ij}\}$
 - 4: Define public whipped map P^* as above
 - 5: **Output:** secret key (F, T) ; public key P^* (with parameters n, m, o, k, q and $\{E_{ij}\}$)
-

3.4.9 Signing Procedure

Algorithm 5 Procedure: MAYO Signing

- 1: **Input:** message M , secret key (F, T)
 - 2: $\mathbf{h} \leftarrow H(M) \in \mathbb{F}_q^m$
 - 3: **repeat**
 - 4: **for** $i = 1, \dots, k$ **do**
 - 5: Sample vinegar vector $\mathbf{v}_i \xleftarrow{\$} \mathbb{F}_q^v$
 - 6: Form $x_i = (\mathbf{v}_i, \mathbf{o}_i) \in \mathbb{F}_q^n$ (concatenation of vinegar and oil variables, with $\mathbf{o}_i \in \mathbb{F}_q^o$)
 - 7: **end for**
 - 8: Substitute $\{x_i\}$ into P^* ; obtain an $m \times (ko)$ linear system in $\{\mathbf{o}_i\}$
 - 9: Solve for $\{\mathbf{o}_i\}$; if unsolvable, repeat with new $\{\mathbf{v}_i\}$
 - 10: **until** system is solvable
 - 11: **Output:** signature $(x_1, \dots, x_k)$ (or compressed form)
-

3.4.10 Verification Procedure

Algorithm 6 Procedure: MAYO Verification

- 1: **Input:** message M , signature $(x_1, \dots, x_k) \in (\mathbb{F}_q^n)^k$, public key P^*
 - 2: Check that each $x_i \in \mathbb{F}_q^n$; if invalid, **return** Reject
 - 3: $\mathbf{h} \leftarrow H(M) \in \mathbb{F}_q^m$
 - 4: Compute $\mathbf{h}' \leftarrow P^*(x_1, \dots, x_k)$
 - 5: **Return** Accept if $\mathbf{h}' = \mathbf{h}$; otherwise, Reject
-

3.4.11 Complexity

Verification requires evaluation of m quadratic polynomials in kn variables ($O(m(kn)^2)$ operations). Signing involves solving an $m \times (k \cdot o)$ linear system, which is efficient provided $k \cdot o \geq m$. Key generation is dominated by sampling affine transformations and defining emulsifiers.

3.4.12 Security Summary

MQ hardness. Solving $P^*(x_1, \dots, x_k) = h$ is difficult under the MQ assumptions. **Indistinguishability.** The whipped structure with emulsifiers hides the reduced oil dimension, making the public map computationally indistinguishable from random quadratics [Wol05]. **Attacks.** Gröbner basis, MinRank, and differential attacks are addressed by ensuring $ko \geq m$ and using secure emulsifier designs [Beu21, Hei25]. Grover provides a quadratic speedup; parameters are chosen conservatively to retain NIST-level security.

3.4.13 Parameter Selection (Practitioner Notes)

- Choose o small for compactness, but ensure that $ko \geq m$ is solvable.
- Emulsifier matrices must be fixed globally and selected to prevent algebraic leakage.
- The size of the signature increases with k ; public keys remain compact relative to UOV.
- Use recommended parameter sets (e.g., MAYO-1/3/5) from [Beu21].

4 Our Construction

Adaptor signatures provide conditional signing capabilities that are critical in advanced cryptographic protocols such as atomic swaps, fair exchange, and privacy-preserving contracts [PBF⁺18]. Designing such constructions secure against quantum adversaries is essential in the post-quantum setting. Multivariate schemes, particularly UOV [KPG99] and MAYO [Beu21], are attractive candidates due to their efficiency and security, which stem from the hardness of the MQ problem [Wol05].

We propose a *multivariate witness-hiding adaptor signature* (MWAS) scheme that generically extends these base schemes with adaptor functionality.

Design Approach (Commit-then-Sign). Our construction follows a *commit-then-sign* paradigm rather than the classical algebraic approach where the witness is embedded into signing randomness. This design choice is deliberate: multivariate quadratic signature schemes such as UOV and MAYO do not expose signing randomness in a form that permits algebraic witness embedding. Instead, MWAS commits to the witness via an HMAC-SHA256 commitment, incorporates the commitment into the signed message, and appends the witness to form the complete signature. Extraction is therefore direct (by parsing the concatenated witness) and relies on the binding property of the commitment for security.

This differs from algebraic adaptor signatures (e.g., Schnorr-based schemes [PBF⁺18]) and from the generic framework of Gerhart et al. [GSST24], which characterise adaptor signatures in terms of pre-verify soundness, malleability resistance, and leakage-freeness. Our construction satisfies pre-verify soundness (Lemma 4) and achieves witness hiding and extractability through the PRF and binding properties of HMAC-SHA256. The absence of algebraic embedding means that witness-programmability (i.e., the ability to adapt a pre-signature to an arbitrary chosen witness) is limited to the hash-preimage relation rather than arbitrary NP relations. We make this limitation explicit in Section 7.

The central idea is to link an external witness value to a signature through a lightweight *commit-and-verify* mechanism. In particular, we employ an HMAC-SHA256-based commitment [KBC97, GH03] to derive a statement from the witness, and a concatenation-based adaptation that transforms pre-signatures into complete signatures once the witness is revealed. This enables the efficient extraction of the witness from any valid pair of pre-signatures and complete signatures.

Unlike more complex algebraic frameworks for adaptor signatures [PBF⁺18], our approach deliberately favours simplicity: relying on canonical concatenation rather than non-black-box transformations. This makes the scheme practical, efficient, and fully compatible with existing multivariate implementations, while preserving the desired adaptor signature properties of correctness, adaptability, extractability, and unforgeability.

Witness Sizes. In MWAS, the witness size depends on the underlying base scheme and the security level. Table 1 summarises the concrete witness lengths for UOV and MAYO at

Table 1: Comparison of witness sizes (in bytes and bits) for UOV and MAYO across NIST security levels.

Scheme Variant	Target Security	Witness Length
UOV-128 (OV-Is)	Level I (128-bit)	48 B (384 bits)
UOV-192 (OV-Ip)	Level III (192-bit)	64 B (512 bits)
UOV-256 (OV-III)	Level V (256-bit)	80 B (640 bits)
MAYO-128 (MAYO-1)	Level I (128-bit)	24 B (192 bits)
MAYO-192 (MAYO-3)	Level III (192-bit)	32 B (256 bits)
MAYO-256 (MAYO-5)	Level V (256-bit)	40 B (320 bits)

the NIST security levels. These values follow the parameterisations defined in the original proposals [KPG99, Beu21].

Commitment and Relation. To bind a witness w to a public statement Y , MWAS employs a deterministic SHA256-based key derivation followed by an HMAC commitment [BCC⁺24]. Formally, for a witness $w \in \mathcal{W}$, we define:

$$\begin{aligned}
k &:= \text{SHA256}(\langle \text{ADAPTOR_COMMITMENT_SALT_v1} \rangle \parallel \langle w \rangle), \\
h_{\text{com}} &:= \text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \parallel \langle w \rangle), \\
Y &:= k \parallel h_{\text{com}}.
\end{aligned}$$

The induced relation $\mathcal{R}$ is the following:

$$\begin{aligned}
\mathcal{R} = \left\{ (w, Y) \mid Y = (k, h_{\text{com}}) \text{ where} \right. \\
\quad k = \text{SHA256}(\langle \text{ADAPTOR_COMMITMENT_SALT_v1} \rangle \parallel \langle w \rangle), \\
\quad \left. h_{\text{com}} = \text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \parallel \langle w \rangle) \right\}.
\end{aligned}$$

Encoding Convention. The notation $\langle \cdot \rangle$ denotes canonical length-prefixed encoding of bitstrings: $\langle s \rangle = \text{len}(s) \parallel s$, where $\text{len}(s)$ is a fixed-width (4-byte big-endian) encoding of the byte length of s . This ensures unambiguous concatenation and avoids parsing ambiguities.

Domain Separation. The concrete domain-separation labels used in this section (`ADAPTOR_COMMITMENT_SALT_v1` for key derivation and `ADAPTORv1` for the HMAC tag) correspond to the abstract labels "STATEMENT" and "BINDING" used in the security model of Section 5. In both cases, domain separation prevents oracle collisions across different uses of the hash function. A full mapping is provided in Section 5.1.

Definition. MWAS is characterised by a family of (PPT) algorithms, described as follows:

$$\text{MWAS} = (\text{KeyGen}, \text{GenerateStatement}, \text{PreSign}, \text{PreVerify}, \text{Adapt}, \text{Verify}, \text{Extract}),$$

where:

- $\text{KeyGen}(1^\lambda) \rightarrow (pk, sk)$: Generates a key pair consisting of a public verification key pk and a private signing key sk , based on the underlying multivariate scheme Σ at the security parameter $\lambda \in \{128, 192, 256\}$.
- $\text{GenerateStatement}(pk, w) \rightarrow Y$: Produces a public statement Y corresponding to a witness w , using the commitment mechanism associated with the relationship $\mathcal{R}$. The public key pk is used to enforce the witness length check via `WitnessLenOK`.

Algorithm 7 MWAS Construction

```

1: KEYGEN( $\lambda$ )
   Ensure:  $(pk, sk)$ 
2:   if  $\lambda \notin \{128, 192, 256\}$  then
3:     return  $\perp$ 
4:   end if
5:    $(pk, sk) \leftarrow \Sigma.\text{KeyGen}(1^\lambda)$ 
6:   if  $pk = \perp \vee sk = \perp$  then
7:     return  $\perp$ 
8:   end if
9:   return  $(pk, sk)$ 
10: end

11: GENERATESTATEMENT( $pk, w$ )
   Ensure:  $Y$ 
12:   if  $w = \perp \vee \neg \text{WitnessLenOK}(pk, |w|)$  then
13:     return  $\perp$ 
14:   end if
15:    $k \leftarrow \text{SHA256}(\langle \text{ADAPTOR\_COMMITMENT\_SALT\_v1} \rangle \| \langle w \rangle)$ 
16:    $h_{\text{com}} \leftarrow \text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \| \langle w \rangle)$ 
17:    $Y \leftarrow (k, h_{\text{com}})$   $\triangleright |Y| = 512$  bits
18:   return  $Y$ 
19: end

20: PRESIGN( $sk, m, Y$ )
   Ensure:  $\hat{\sigma}$ 
21:   if  $sk = \perp \vee m = \perp \vee Y = \perp \vee |Y| \neq 512$  then
22:     return  $\perp$ 
23:   end if
24:    $m' \leftarrow \langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle$ 
25:    $h_m \leftarrow \text{SHA256}(m')$ 
26:    $\sigma' \leftarrow \Sigma.\text{Sign}(sk, h_m)$ 
27:   if  $\sigma' = \perp$  then
28:     return  $\perp$ 
29:   end if
30:   return  $\hat{\sigma} = (\sigma', Y, h_m)$ 
31: end

32: PREVERIFY( $pk, m, \hat{\sigma}$ )
   Ensure: 0/1
33:   if  $pk = \perp \vee m = \perp \vee \hat{\sigma} = \perp$  then
34:     return 0
35:   end if
36:   parse  $\hat{\sigma} = (\sigma', Y, h_m)$ 
37:   if  $Y = \perp \vee |Y| \neq 512$  then
38:     return 0
39:   end if
40:    $m' \leftarrow \langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle$ 
41:    $h'_m \leftarrow \text{SHA256}(m')$ 
42:   if  $h_m \neq h'_m \vee \sigma' = \perp$  then
43:     return 0
44:   end if
45:   return  $\Sigma.\text{Verify}(pk, h_m, \sigma')$ 
46: end

53: ADAPT( $pk, m, \hat{\sigma}, w$ )
   Ensure:  $\sigma$ 
54:   if  $\hat{\sigma} = \perp \vee w = \perp \vee \neg \text{WitnessLenOK}(pk, |w|)$ 
   then
55:     return  $\perp$ 
56:   end if
57:   if  $\text{PREVERIFY}(pk, m, \hat{\sigma}) \neq 1$  then
58:     return  $\perp$ 
59:   end if
60:   parse  $\hat{\sigma} = (\sigma', Y, h_m)$  and  $Y = (k, h_{\text{com}})$ 
61:   if  $Y = \perp \vee |Y| \neq 512 \vee \sigma' = \perp$  then
62:     return  $\perp$ 
63:   end if
64:   if  $\text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \| \langle w \rangle) \neq h_{\text{com}}$ 
   then
65:     return  $\perp$ 
66:   end if
67:   return  $\sigma = \langle \sigma' \rangle \| \langle w \rangle$ 
68: end

69: VERIFY( $pk, m, \sigma$ )
   Ensure: 0/1
70:   if  $pk = \perp \vee m = \perp \vee \sigma = \perp$  then
71:     return 0
72:   end if
73:   parse  $\sigma = \langle \sigma' \rangle \| \langle w \rangle$ 
74:   if  $w = \perp \vee \neg \text{WitnessLenOK}(pk, |w|)$  then
75:     return 0
76:   end if
77:    $Y \leftarrow \text{GENERATESTATEMENT}(pk, w)$ 
78:   if  $Y = \perp \vee |Y| \neq 512$  then
79:     return 0
80:   end if
81:    $m' \leftarrow \langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle$ 
82:    $h_m \leftarrow \text{SHA256}(m')$ 
83:   return  $\Sigma.\text{Verify}(pk, h_m, \sigma')$ 
84: end

85: EXTRACT( $\hat{\sigma}, \sigma$ )
   Ensure:  $w$ 
86:   if  $\hat{\sigma} = \perp \vee \sigma = \perp$  then
87:     return  $\perp$ 
88:   end if
89:   parse  $\hat{\sigma} = (\sigma', Y, h_m)$  and  $\sigma = \langle \sigma'' \rangle \| \langle w \rangle$ 
90:   if  $w = \perp \vee |Y| \neq 512$  then
91:     return  $\perp$ 
92:   end if
93:   if  $\sigma' \neq \sigma''$  then
94:     return  $\perp$ 
95:   end if
96:   parse  $Y = (k, h_{\text{com}})$ 
97:   if  $\text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \| \langle w \rangle) \neq h_{\text{com}}$ 
   then
98:     return  $\perp$ 
99:   end if
100:  return  $w$ 
101: end

```

- $\text{PreSign}(sk, m, Y) \rightarrow \hat{\sigma}$: With secret key sk , message m , and statement Y , this procedure outputs a pre-signature $\hat{\sigma}$ binding m to Y .
- $\text{PreVerify}(pk, m, \hat{\sigma}) \rightarrow \{0, 1\}$: Verifies whether $\hat{\sigma}$ is a valid pre-signature for m under the public key pk .
- $\text{Adapt}(pk, m, \hat{\sigma}, w) \rightarrow \sigma$: Converts a valid pre-signature $\hat{\sigma}$ into a complete signature σ using witness w , provided that $(w, Y) \in \mathcal{R}$. The public key pk and message m are used internally by PreVerify to validate the pre-signature before adaptation.
- $\text{Verify}(pk, m, \sigma) \rightarrow \{0, 1\}$: Checks if σ constitutes a valid signature on m relative to pk .
- $\text{Extract}(\hat{\sigma}, \sigma) \rightarrow w$: From a valid pre-signature $\hat{\sigma}$ and its completed signature σ on the same message, recovers the unique witness w with $(w, Y) \in \mathcal{R}$; outputs $\perp$ if no valid witness exists.

5 Security Properties and Proofs

This section formally examines the security guarantees of MWAS. The construction inherits its security from three well-established assumptions: (i) the existential unforgeability under chosen-message attacks (EUF-CMA) of the base scheme $\Sigma \in \{\text{UOV}, \text{MAYO}\}$, (ii) the PRF security of HMAC-SHA256 [BCK96b], and (iii) the collision resistance of SHA-256, where SHA models SHA-256 as a random oracle [CGH04]. Domain separation is enforced via fixed labels to prevent oracle collisions.

We focus on the following security properties, which together characterise the soundness of adaptor signatures:

- **Correctness:** any honestly generated pre-signature and its adapted completion are always accepted by the verification algorithm.
- **Witness Hiding:** given only a pre-signature and its statement, no efficient adversary can extract more than negligible information about the witness.
- **Witness Recoverability:** a valid pre-signature together with the corresponding completed signature uniquely determines the underlying witness and allows its efficient recovery.
- **Witness Uniqueness:** every valid pre-signature can be completed with at most one witness, ruling out substitution attacks.
- **Pre-signature Privacy:** pre-signatures derived from different witnesses are computationally indistinguishable to outside parties.
- **Adaptation Soundness:** combining a valid pre-signature with a valid witness always produces an accepting full signature.
- **Unforgeability:** no PPT adversary can forge a new pre-signature on an unqueried message with non-negligible success.
- **Forward Security:** even if a witness is later revealed, past pre-signatures remain secure and unexploitable.
- **Statement Binding:** each valid statement is associated with a single admissible witness, preventing ambiguity in commitments.

In addition to the ROM, we analyse security in the *Quantum Random Oracle Model* (QROM) [HXY19], which captures post-quantum adversaries issuing superposition queries to the hash oracle. We use standard game-based definitions and show that breaking MWAS implies either forging the base signature scheme or violating HMAC security.

5.1 Security Model and Assumptions

We fix the security parameter $\lambda \in \{128, 192, 256\}$ and assume that all algorithms run in probabilistic polynomial time in λ . Security is analysed in the ROM, where SHA is modelled as a random oracle. In the post-quantum setting, we use the QROM, which allows adversaries to issue superposition queries to SHA. Domain separation is enforced through fixed labels to avoid oracle collisions.

Notation and Definitions. Let $\mathcal{W} = \{0, 1\}^*$ denote the witness space (witnesses have scheme-specific lengths enforced by WitnessLenOK; see Table 1) and $\mathcal{Y} = \{0, 1\}^{512}$ denote the statement space (always 512 bits: two SHA-256 outputs). We define the relation $\mathcal{R} \subseteq \mathcal{W} \times \mathcal{Y}$ as:

$$\mathcal{R} = \{(w, Y) : Y = \text{GenerateStatement}(w)\}$$

where $\text{GenerateStatement} : \mathcal{W} \rightarrow \mathcal{Y}$ is defined concretely as in Algorithm 7:

$$k = \text{SHA}(\text{"STATEMENT"} \| w), \quad h_{\text{com}} = \text{HMAC}_k(\text{"BINDING"} \| w), \quad \text{GenerateStatement}(w) = k \| h_{\text{com}}.$$

The abstract labels "STATEMENT" and "BINDING" correspond to the concrete labels ADAPTOR_COMMITMENT_SALT_v1 and ADAPTORv1 respectively. Due to SHA-256 collision resistance and the PRF property of HMAC, this architecture guarantees binding with overwhelming probability.

Adversary Model. Adversaries may be either PPT or QPT. They may issue the following oracle queries:

- $\mathcal{O}_{\text{PreSign}}(m, Y)$: returns $\text{PreSign}(sk, m, Y)$ under MWAS,
- $\mathcal{O}_{\text{Adapt}}(m, \hat{\sigma}, w)$: returns $\text{Adapt}(pk, m, \hat{\sigma}, w)$,
- $\mathcal{O}_{\text{SHA}}(x)$: returns $\text{SHA}(x)$ from the random oracle.

Our security games follow the standard oracle access model. The set of queries made by the adversary is denoted by $\mathcal{Q}$.

Cryptographic Assumptions.

1. **Multivariate Signature Security:** The base scheme $\Sigma \in \{\text{UOV}, \text{MAYO}\}$ is EUF-CMA secure in the ROM (resp. QROM). This implies that for every PPT or QPT adversary $\mathcal{A}$:

$$\text{Adv}_{\Sigma, \mathcal{A}}^{\text{EUF-CMA}}(\lambda) \leq \text{negl}(\lambda)$$

2. **HMAC Security:** HMAC-SHA256 is a secure (PRF). For any PPT adversary $\mathcal{A}$ making at most q queries:

$$\text{Adv}_{\text{HMAC}, \mathcal{A}}^{\text{PRF}}(\lambda) \leq \text{negl}(\lambda)$$

3. **Collision Resistance:** SHA-256 is collision-resistant. For any PPT adversary $\mathcal{A}$:

$$\text{Adv}_{\text{SHA}, \mathcal{A}}^{\text{CR}}(\lambda) \leq \text{negl}(\lambda)$$

4. **Multivariate Problem Hardness:** The underlying multivariate quadratic (MQ) problem is hard. For UOV, this is the MQ problem over $\mathbb{F}_q$; for MAYO, this is the MQ problem with additional structure. For any PPT adversary $\mathcal{A}$:

$$\text{Adv}_{\mathcal{A}}^{\text{MQ}}(\lambda) \leq \text{negl}(\lambda)$$

Security Parameterization. The witness space $\mathcal{W} = \{0,1\}^*$ and the statement space $\mathcal{Y} = \{0,1\}^{512}$ are chosen to ensure:

- **Binding:** Because of collision resistance, there is very little chance that two distinct witnesses will map to the same statement.
- **Extractability:** It is highly likely that the witness can be effectively retrieved from the statement.
- **Hiding:** No information about the witness is revealed by the statement alone.

Domain Separation. We use the following fixed labels for domain separation:

- "ADAPTORv1": For general adaptor signature operations
- "PRESIGN": For pre-signature generation
- "STATEMENT": For statement generation (key derivation)
- "BINDING": For binding commitment (HMAC tag)

Security Bounds. For every adversary $\mathcal{A}$ attacking MWAS, efficient reductions exist. There exist reducers $\mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}$ such that:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{aEUF-CMA}}(\lambda) \leq \text{Adv}_{\Sigma, \mathcal{B}}^{\text{EUF-CMA}}(\lambda) + \text{negl}(\lambda) \quad (1)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) \leq \text{Adv}_{\text{HMAC}, \mathcal{C}}^{\text{PRF}}(\lambda) + \text{negl}(\lambda) \quad (2)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{EXT}}(\lambda) \leq \text{Adv}_{\text{SHA}, \mathcal{D}}^{\text{CR}}(\lambda) + \text{negl}(\lambda) \quad (3)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{BIND}}(\lambda) \leq \text{Adv}_{\text{SHA}, \mathcal{E}}^{\text{CR}}(\lambda) + \text{negl}(\lambda) \quad (4)$$

These bounds show that MWAS inherits its security from the underlying assumptions. The reductions are constructed in the subsequent proof sections, where we show how to transform any successful MWAS adversary into an adversary against one of the base assumptions.

Security Summary. MWAS provides the following security guarantees:

- **Unforgeability:** Inherited from the base multivariate signature scheme
- **Witness Hiding:** Based on HMAC PRF security
- **Witness Extractability:** Based on SHA-256 collision-resistance
- **Statement Binding:** Based on SHA-256 collision-resistance
- **Forward Security:** Inherited from the base scheme's security

All security properties hold in both the classical ROM and the post-quantum QROM, making MWAS suitable for post-quantum applications.

5.2 Formal Definitions

We formalise the security properties of MWAS. Let

$$\text{MWAS} = (\text{KeyGen}, \text{GenerateStatement}, \text{PreSign}, \text{PreVerify}, \text{Adapt}, \text{Verify}, \text{Extract})$$

denote the tuple of algorithms. All adversaries are assumed to operate in either PPT or QPT. All hash functions are modelled as random oracles. Let $\mathcal{R}$ be the relation defined in Section 5.1, and let $\mathcal{Q}$ denote the set of queries made by the adversary to the oracles.

Definition 1 (Correctness). An MWAS scheme satisfies *correctness* if, for every security parameter λ , any key pair $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$, any message m , and any witness-statement pair $(w, Y) \in \mathcal{R}$, the following holds:

$$\begin{aligned} \hat{\sigma} &\leftarrow \text{PreSign}(sk, m, Y), \\ \sigma &\leftarrow \text{Adapt}(pk, m, \hat{\sigma}, w) \Rightarrow \text{Verify}(pk, m, \sigma) = 1 \wedge \text{Extract}(\hat{\sigma}, \sigma) = w. \end{aligned}$$

Definition 2 (Completeness (COMP)). An MWAS scheme satisfies *completeness* if, for every security parameter λ , every key pair $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$, and all pairs $(w, Y) \in \mathcal{R}$, the following holds with all but negligible probability:

$$\begin{aligned} \hat{\sigma} &\leftarrow \text{PreSign}(sk, m, Y), \quad \sigma \leftarrow \text{Adapt}(pk, m, \hat{\sigma}, w) \\ \Rightarrow \quad &\text{PreVerify}(pk, m, \hat{\sigma}) = 1 \wedge \text{Verify}(pk, m, \sigma) = 1. \end{aligned}$$

Definition 3 (Pre-Verification Soundness (SOUND)). An MWAS scheme is called *sound* if any probabilistic (classical or quantum) adversary $\mathcal{A}$ can produce only with negligible probability a triple $(m, Y, \hat{\sigma})$ that passes the pre-verification check but admits no valid completion. Formally:

$$\Pr \left[\text{PreVerify}(pk, m, \hat{\sigma}) = 1 \wedge \forall w \in \mathcal{W} : \text{Verify}(pk, m, \text{Adapt}(pk, m, \hat{\sigma}, w)) = 0 \right] \leq \text{negl}(\lambda).$$

Definition 4 (Witness Hiding (WH)). An MWAS construction provides *witness hiding* if no efficient adversary can distinguish which of two witnesses was used in the creation of a pre-signature, beyond negligible probability.

Formally, for any PPT or QPT adversary $\mathcal{A}$ with oracle access to $\mathcal{O}_{\text{PreSign}}(m', Y')$ for $m' \neq m$, we require

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) = \left| \Pr [\mathcal{A}^{\mathcal{O}_{\text{PreSign}}}(pk, Y_0, \hat{\sigma}_0) = 1] - \Pr [\mathcal{A}^{\mathcal{O}_{\text{PreSign}}}(pk, Y_1, \hat{\sigma}_1) = 1] \right| \leq \text{negl}(\lambda),$$

where for $b \in \{0, 1\}$ we set $Y_b \leftarrow \text{GenerateStatement}(w_b)$ and $\hat{\sigma}_b \leftarrow \text{PreSign}(sk, m, Y_b)$ with $|w_0| = |w_1|$ and $(w_b, Y_b) \in \mathcal{R}$.

Definition 5 (Witness Recovery Property (EXT)). We say that an MWAS construction satisfies *witness recovery* if there exists a PPT algorithm (the extractor $\mathcal{E}$) such that, for every PPT/QPT adversary $\mathcal{A}$ with oracle access to $\mathcal{O}_{\text{PreSign}}$ and $\mathcal{O}_{\text{Adapt}}$, the following holds.

Whenever $\mathcal{A}$ outputs a pair $(\hat{\sigma}, \sigma)$ that is accepted as a valid pre-signature and signature on some message m , the extractor succeeds in returning a witness w with $(w, Y) \in \mathcal{R}$ except with negligible probability. Formally,

$$\Pr[\text{Verify}(pk, m, \sigma) = 1 \wedge \mathcal{E}(\hat{\sigma}, \sigma) = \perp] \leq \text{negl}(\lambda).$$

Definition 6 (Witness Uniqueness (WU)). An MWAS construction achieves *witness uniqueness* if, except with negligible probability, a valid pre-signature $\hat{\sigma}$ for a message m and statement Y admits at most one completing witness. More precisely, for any two distinct witnesses $w \neq w'$,

$$\Pr[\text{Verify}(pk, m, \text{Adapt}(pk, m, \hat{\sigma}, w)) = 1 \wedge \text{Verify}(pk, m, \text{Adapt}(pk, m, \hat{\sigma}, w')) = 1] \leq \text{negl}(\lambda).$$

Definition 7 (Statement Binding (BIND)). MWAS is *statement binding* if no PPT/QPT adversary $\mathcal{A}$ can produce distinct witnesses $w \neq w'$ that bind to the same statement Y :

$$\Pr[(w, Y) \in \mathcal{R} \wedge (w', Y) \in \mathcal{R} \wedge w \neq w'] \leq \text{negl}(\lambda),$$

where (w, w', Y) is output by $\mathcal{A}$.

Definition 8 (Pre-signature Privacy (PSP)). An MWAS scheme achieves *pre-signature privacy* if observing pre-signatures does not allow an efficient adversary to tell which witness was used, beyond a negligible advantage. Formally, for any PPT/QPT adversary $\mathcal{A}$,

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{PSP}}(\lambda) = \left| \Pr[\mathcal{A}^{\mathcal{O}_{\text{PreSign}}}(pk, \hat{\sigma}_0) = 1] - \Pr[\mathcal{A}^{\mathcal{O}_{\text{PreSign}}}(pk, \hat{\sigma}_1) = 1] \right| \leq \text{negl}(\lambda),$$

where for $b \in \{0, 1\}$, the pre-signature $\hat{\sigma}_b$ is generated as $\hat{\sigma}_b \leftarrow \text{PreSign}(sk, m, Y_b)$ with $Y_b \leftarrow \text{GenerateStatement}(w_b)$ and $(w_b, Y_b) \in \mathcal{R}$.

Remark 1 (Relation between Witness Hiding and Pre-Signature Privacy). Definition 4 (Witness Hiding) gives the adversary the statement Y , whereas Definition 8 (Pre-Signature Privacy) does not. Therefore Witness Hiding is formally the stronger notion. In our MWAS construction, however, the pre-signature $\hat{\sigma} = (\sigma', Y, h_m)$ already contains the statement Y , so any adversary in the Pre-Signature Privacy game can immediately read Y from $\hat{\sigma}$. Consequently, the two notions are equivalent in our construction.

Definition 9 (Adaptation Soundness (AS)). An MWAS scheme ensures *adaptation soundness* if no efficient adversary can produce a pre-signature and a corresponding witness that pass the pre-verification check, but fail to yield an accepting final signature. Formally, for any PPT/QPT adversary $\mathcal{A}$,

$$\Pr[\text{PreVerify}(pk, m, \hat{\sigma}) = 1 \wedge (w, Y) \in \mathcal{R} \wedge \text{Verify}(pk, m, \text{Adapt}(pk, m, \hat{\sigma}, w)) = 0] \leq \text{negl}(\lambda),$$

where $(\hat{\sigma}, w)$ is generated by $\mathcal{A}$.

Definition 10 (Adaptive Unforgeability (aEUF-CMA)). An MWAS construction achieves *adaptive existential unforgeability* if no PPT or QPT adversary $\mathcal{A}$, even when given oracle access to $\mathcal{O}_{\text{PreSign}}$ and $\mathcal{O}_{\text{Adapt}}$, can produce a valid signature on a message that was never submitted to either oracle. Formally,

$$\Pr[\text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin \mathcal{Q}] \leq \text{negl}(\lambda),$$

where $(\hat{\sigma}^*, \sigma^*)$ is the adversary's output and $\mathcal{Q}$ is the set of queries made to the oracles.

Definition 11 (Pre-signature Unforgeability (PreEUF-CMA)). An MWAS scheme is said to provide *pre-signature unforgeability* if no polynomial-time (classical or quantum) adversary $\mathcal{A}$, even with access to the $\mathcal{O}_{\text{PreSign}}$ and $\mathcal{O}_{\text{Adapt}}$ oracles, can successfully fabricate a pre-signature on a message that was never submitted to the pre-signing oracle. Formally,

$$\Pr[\text{PreVerify}(pk, m^*, \hat{\sigma}^*) = 1 \wedge m^* \notin \mathcal{Q}] \leq \text{negl}(\lambda),$$

where $\hat{\sigma}^*$ denotes the adversary's output and $\mathcal{Q}$ is the set of messages queried to $\mathcal{O}_{\text{PreSign}}$.

Definition 12 (Forward Security (FS)). An MWAS construction achieves *forward security under witness exposure* if, even after the adversary $\mathcal{A}$ learns a valid witness w corresponding to some statement, it remains computationally infeasible to forge new signatures on previously unseen messages. Formally, for any PPT/QPT adversary with access to $\mathcal{O}_{\text{PreSign}}$,

$$\Pr[\text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin \mathcal{Q}] \leq \text{negl}(\lambda),$$

where σ^* is the adversary's output after witness exposure, and $\mathcal{Q}$ denotes the set of messages previously queried to $\mathcal{O}_{\text{PreSign}}$.

Definition 13 (Zero-Knowledge for the Witness (ZK)). An MWAS scheme is said to provide *zero-knowledge for the witness* if there exists a PPT simulator $\mathcal{S}$ such that for every PPT/QPT distinguisher $\mathcal{D}$, the distributions of honestly generated pre-signatures and simulated pre-signatures are computationally indistinguishable. Formally,

$$\text{Adv}_{\text{MWAS}, \mathcal{D}}^{\text{ZK}}(\lambda) = \left| \Pr [\mathcal{D}(pk, m, Y, \hat{\sigma}) = 1] - \Pr [\mathcal{D}(pk, m, Y, \hat{\sigma}_{\text{sim}}) = 1] \right| \leq \text{negl}(\lambda),$$

where $\hat{\sigma} \leftarrow \text{PreSign}(sk, m, Y)$ is an honestly generated pre-signature, and $\hat{\sigma}_{\text{sim}} \leftarrow \mathcal{S}(pk, m, Y)$ is produced by the simulator.

5.3 Security Games

We now formalise the security notions through game-based experiments between a challenger and an adversary $\mathcal{A}$. All experiments are defined in the ROM [BR93]. An adversary wins if it produces an output satisfying the game's winning condition. The adversary's advantage is the probability of winning, minus the trivial baseline (if applicable). We denote by $\mathcal{Q}$ the set of queries made by the adversary to the oracles.

5.3.1 Correctness Property

Correctness. An MWAS scheme is correct if, for every security parameter λ , all key pairs $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$, any message m , and every witness-statement pair (w, Y) with $Y = \text{GenerateStatement}(w)$, we have:

$$\hat{\sigma} \leftarrow \text{PreSign}(sk, m, Y), \quad \sigma \leftarrow \text{Adapt}(pk, m, \hat{\sigma}, w), \quad w' \leftarrow \text{Extract}(\hat{\sigma}, \sigma).$$

Correctness requires that, except with negligible probability,

$$\text{Verify}(pk, m, \sigma) = 1 \quad \text{and} \quad w' = w.$$

5.3.2 Witness Hiding Game

Witness Hiding Experiment $\text{Exp}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(1^\lambda)$

1. The challenger executes $\text{KeyGen}(1^\lambda)$ to generate (pk, sk) and provides pk to the adversary $\mathcal{A}$.
2. The adversary chooses (m, w_0, w_1) such that $(w_0, Y_0), (w_1, Y_1) \in \mathcal{R}$ with $|w_0| = |w_1|$.
3. $\mathcal{A}$ may query signing oracles $\mathcal{O}_{\text{PreSign}}(m', Y')$ on any $m' \neq m$.
4. The challenger selects $b \xleftarrow{\$} \{0, 1\}$, computes $Y_b \leftarrow \text{GenerateStatement}(w_b)$ and $\hat{\sigma} \leftarrow \text{PreSign}(sk, m, Y_b)$, then returns $(\hat{\sigma}, Y_b)$ to $\mathcal{A}$.
5. Finally, $\mathcal{A}$ outputs a guess $b' \in \{0, 1\}$.

Success Criterion: The adversary succeeds if $b' = b$.

Distinguishing Advantage:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

5.3.3 Witness Extractability Game

Witness Extractability Experiment $\text{Exp}_{\text{MWAS}, \mathcal{A}}^{\text{EXT}}(1^\lambda)$

1. The challenger executes $\text{KeyGen}(1^\lambda)$ to obtain a key pair (pk, sk) and provides the public key pk to the adversary $\mathcal{A}$.
2. The adversary is given oracle access and may query:
 - $\mathcal{O}_{\text{PreSign}}(m, Y)$, which returns a pre-signature $\hat{\sigma}$,
 - $\mathcal{O}_{\text{Adapt}}(m, \hat{\sigma}, w)$, which outputs the corresponding full signature σ .
3. Eventually, $\mathcal{A}$ produces a candidate triple $(m^*, \hat{\sigma}^*, \sigma^*)$.
4. The challenger computes $w^* \leftarrow \text{Extract}(\hat{\sigma}^*, \sigma^*)$.

Success Criterion: The adversary succeeds if $\text{Verify}(pk, m^*, \sigma^*) = 1$ but the extractor fails, i.e., $w^* = \perp$.

Extraction Advantage:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{EXT}}(\lambda) = \Pr [\text{Verify}(pk, m^*, \sigma^*) = 1 \wedge w^* = \perp].$$

5.3.4 Unforgeability Challenge (aEUF-CMA)

Definition. The adaptive existential unforgeability experiment $\text{Game}_{\text{MWAS}, \mathcal{A}}^{\text{aEUF-CMA}}(1^\lambda)$ proceeds as follows:

1. A challenger generates a key pair $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$ and gives the public key pk to the adversary $\mathcal{A}$.
2. $\mathcal{A}$ may interact with the system through two interfaces:
 - *Pre-signing oracle:* on input (m, Y) it returns a pre-signature $\hat{\sigma}$.
 - *Adaptation oracle:* on input $(m, \hat{\sigma}, w)$ it returns the corresponding signature σ .
3. After making a sequence of queries, $\mathcal{A}$ outputs a candidate forgery, either a pre-signature $(m^*, \hat{\sigma}^*)$ or a full signature (m^*, σ^*) .

Winning Condition. The adversary is successful if the output is accepted by Verify on some message m^* that was never submitted to either oracle during the game.

Advantage.

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{aEUF-CMA}}(\lambda) = \Pr[\text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin \mathcal{Q}].$$

5.3.5 Witness Uniqueness Challenge

Definition. The witness uniqueness experiment $\text{Game}_{\text{MWAS}, \mathcal{A}}^{\text{WU}}(1^\lambda)$ is defined as follows:

1. The challenger executes $\text{KeyGen}(1^\lambda)$ to obtain (pk, sk) and provides the public key pk to the adversary $\mathcal{A}$.
2. The adversary may interact with the challenger using:
 - A *pre-signature oracle* $\mathcal{O}_{\text{PreSign}}(m, Y)$ that returns a pre-signature $\hat{\sigma}$.
 - An *adaptation oracle* $\mathcal{O}_{\text{Adapt}}(m, \hat{\sigma}, w)$ that returns a full signature σ .
3. Finally, $\mathcal{A}$ outputs $(m, \hat{\sigma}, \sigma, \sigma')$, attempting to demonstrate two distinct witnesses $w \neq w'$ that both lead to valid completions of the same pre-signature.

Winning Condition. The adversary succeeds if both $\sigma = \text{Adapt}(pk, m, \hat{\sigma}, w)$ and $\sigma' = \text{Adapt}(pk, m, \hat{\sigma}, w')$ verify under pk for $w \neq w'$.

Advantage.

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WU}}(\lambda) = \Pr[\text{Verify}(pk, m, \sigma) = 1 \wedge \text{Verify}(pk, m, \sigma') = 1 \wedge w \neq w'].$$

5.3.6 Pre-signature Unforgeability Challenge

Definition. The experiment $\text{Game}_{\text{MWAS}, \mathcal{A}}^{\text{PreEUF-CMA}}(1^\lambda)$ is conducted as follows:

1. The challenger runs $\text{KeyGen}(1^\lambda)$, producing (pk, sk) , and sends the public key pk to the adversary $\mathcal{A}$.
2. The adversary may query two interfaces:
 - $\mathcal{O}_{\text{PreSign}}(m, Y)$, which returns a pre-signature $\hat{\sigma}$.
 - $\mathcal{O}_{\text{Adapt}}(m, \hat{\sigma}, w)$, which returns a corresponding full signature σ .
3. At the end of the interaction, $\mathcal{A}$ outputs a candidate pair $(m^*, \hat{\sigma}^*)$.

Success Criterion. The adversary is said to succeed if m^* was never submitted to $\mathcal{O}_{\text{PreSign}}$ yet $\text{PreVerify}(pk, m^*, \hat{\sigma}^*) = 1$.

Forgery Advantage.

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{PreEUF-CMA}}(\lambda) = \Pr[\text{PreVerify}(pk, m^*, \hat{\sigma}^*) = 1 \wedge m^* \notin \mathcal{Q}].$$

5.3.7 Statement Binding Property

Statement Binding. The construction of MWAS ensures that every valid statement corresponds to a unique witness. Formally, for all security parameters λ and for all witnesses w, w' :

$$Y = \text{GenerateStatement}(w) \wedge Y = \text{GenerateStatement}(w') \Rightarrow w = w'.$$

Binding Condition: With overwhelming probability, no two distinct witnesses can generate the same statement.

5.3.8 Pre-signature Privacy Game

Experiment: Pre-signature Privacy $\text{Exp}_{\text{MWAS}, \mathcal{A}}^{\text{PSP}}(1^\lambda)$

1. The challenger executes $\text{KeyGen}(1^\lambda)$ to obtain (pk, sk) and releases pk to the adversary $\mathcal{A}$.
2. The adversary submits (m, w_0, w_1) such that $(w_0, Y_0), (w_1, Y_1) \in \mathcal{R}$ and $|w_0| = |w_1|$.
3. Oracle queries of the form $\mathcal{O}_{\text{PreSign}}(m', Y')$ are allowed for any $m' \neq m$.
4. The challenger picks a random bit $b \in \{0, 1\}$. Then it computes

$$Y_b \leftarrow \text{GenerateStatement}(w_b), \quad \hat{\sigma}_b \leftarrow \text{PreSign}(sk, m, Y_b),$$

and returns $\hat{\sigma}_b$ to $\mathcal{A}$.

5. The adversary outputs a guess $b' \in \{0, 1\}$.

Success Criterion: The adversary wins if $b' = b$.

Privacy Advantage:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{PSP}}(\lambda) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

5.3.9 Adaptation Soundness Game

Experiment: Adaptation Soundness $\text{Exp}_{\text{MWAS}, \mathcal{A}}^{\text{AS}}(1^\lambda)$

1. The challenger runs $\text{KeyGen}(1^\lambda)$ to obtain a pair (pk, sk) and reveals pk to the adversary $\mathcal{A}$.
2. The adversary is allowed to issue oracle queries to $\mathcal{O}_{\text{PreSign}}(m, Y)$ and $\mathcal{O}_{\text{Adapt}}(m, \hat{\sigma}, w)$.
3. In the end, $\mathcal{A}$ outputs $(m, \hat{\sigma}, w, \sigma)$, where $\hat{\sigma}$ should be a valid pre-signature on m , (w, Y) is a valid relation instance with $Y = \text{GenerateStatement}(w)$, and σ is produced by running $\text{Adapt}(pk, m, \hat{\sigma}, w)$.

Success Criterion: The adversary succeeds if $\text{Verify}(pk, m, \sigma) = 0$ even though $\hat{\sigma}$ and w were valid.

Soundness Advantage:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{AS}}(\lambda) = \Pr [\text{Verify}(pk, m, \sigma) = 0 \wedge \text{PreVerify}(pk, m, \hat{\sigma}) = 1 \wedge (w, Y) \in \mathcal{R}].$$

The scheme is *adaptation sound* if $\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{AS}}(\lambda) \leq \text{negl}(\lambda)$ for all PPT/QPT adversaries $\mathcal{A}$.

5.3.10 Forward Security Game**Experiment: Forward Security** $\text{Exp}_{\text{MWAS}, \mathcal{A}}^{\text{FS}}(1^\lambda)$

1. The challenger runs $\text{KeyGen}(1^\lambda)$ to obtain (pk, sk) and provides pk to the adversary $\mathcal{A}$.
2. The adversary may query the oracle $\mathcal{O}_{\text{PreSign}}(m, Y)$ to obtain pre-signatures on messages of its choice.
3. At some point, the challenger discloses a specific witness w^* to $\mathcal{A}$.
4. Eventually, $\mathcal{A}$ produces a candidate forgery $(m^*, \hat{\sigma}^*)$.

Success Criterion: The adversary is successful if $\hat{\sigma}^*$ corresponds to a valid pre-signature on m^* that was issued prior to the disclosure of w^* .

Forward Security Advantage:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{FS}}(\lambda) = \Pr [\text{PreVerify}(pk, m^*, \hat{\sigma}^*) = 1 \wedge m^* \notin \mathcal{Q}_{\text{after}}] \leq \text{negl}(\lambda).$$

5.4 Theorems and Proof Outlines

We now establish the security assurances of MWAS under the following three assumptions:

1. The foundational multivariate signature scheme $\Sigma \in \{\text{UOV}, \text{MAYO}\}$ is existentially unforgeable under chosen-message attacks (EUF-CMA) within the ROM.
2. HMAC-SHA256 functions as a secure PRF [BCK96b, BCK96a].
3. SHA-256 exhibits collision resistance, with SHA modelled as a random oracle.

5.4.1 Supporting Lemmas

We begin by establishing the following auxiliary properties, which are essential for proving the main theorems.

Lemma 1 (Statement Binding). *If the hash function SHA (modelled here as SHA-256) is collision-resistant, then the procedure GenerateStatement satisfies binding.*

Proof. Assume for contradiction that an adversary $\mathcal{A}$ can find two distinct witnesses $w \neq w'$ that are both associated with the same statement Y . By the definition of GenerateStatement, this requires that

$$Y = \begin{cases} k \parallel \text{HMAC}_k(\text{"BINDING"} \parallel w), & k = \text{SHA}(\text{"STATEMENT"} \parallel w), \\ k' \parallel \text{HMAC}_{k'}(\text{"BINDING"} \parallel w'), & k' = \text{SHA}(\text{"STATEMENT"} \parallel w'). \end{cases}$$

Since both expressions equal Y , their first 256-bit components must satisfy $k = k'$, i.e., $\text{SHA}(\text{"STATEMENT"}\|w) = \text{SHA}(\text{"STATEMENT"}\|w')$. Because $w \neq w'$, this is a collision in SHA.

This contradicts the collision-resistance of SHA-256. Therefore, the probability that $\mathcal{A}$ will break the binding is at most

$$\Pr[\mathcal{A} \text{ breaks binding}] \leq \text{Adv}_{\text{SHA}}^{\text{CR}}(\lambda).$$

□

Lemma 2 (Pre-signature Unforgeability (PreEUF-CMA)). *If the underlying multivariate signature scheme Σ is secure against existential forgery under chosen-message attacks (EUF-CMA), then the adaptor construction MWAS also achieves PreEUF-CMA security.*

Proof. Assume there exists a PPT adversary $\mathcal{A}$ that outputs a valid pre-signature $\hat{\sigma}^*$ on a new message m^* such that $\text{PreVerify}(pk, m^*, \hat{\sigma}^*) = 1$, without having queried m^* to the pre-signing oracle. From this adversary, we can define a reduction $\mathcal{B}$ against the EUF-CMA security of the base scheme Σ : $\mathcal{B}$ interacts with its EUF-CMA challenger, relays oracle queries of $\mathcal{A}$ using the underlying Σ signing oracle, and outputs the embedded base-level forgery contained in $\hat{\sigma}^*$.

Because successful pre-verification requires that the embedded Σ -signature is valid, any forgery by $\mathcal{A}$ immediately yields a forgery against Σ . Hence the advantage of $\mathcal{A}$ is bounded by that of $\mathcal{B}$ in the EUF-CMA game:

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{PreEUF-CMA}}(\lambda) \leq \text{Adv}_{\Sigma, \mathcal{B}}^{\text{EUF-CMA}}(\lambda).$$

□

Lemma 3 (Zero-Knowledge for the Witness). *If HMAC-SHA256 behaves as a PRF, then MWAS achieves zero-knowledge with respect to the witness.*

Proof. Given a target statement $Y = (k, h_{\text{com}})$ and message m , a simulator $\mathcal{S}$ constructs a pre-signature as follows: compute $h_m = \text{SHA}(\langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle)$, invoke the Σ -signing oracle on h_m to obtain σ' , and output $\hat{\sigma}_{\text{sim}} = (\sigma', Y, h_m)$. Because h_{com} is a function of the witness w under HMAC with key k , and HMAC-SHA256 is a secure PRF, the tag h_{com} is computationally indistinguishable from a uniformly random string when k is derived from w via SHA-256. No efficient distinguisher $\mathcal{D}$ can tell whether the h_{com} component was computed from the real witness or is pseudorandom, without knowing w . The base-signature component σ' is identically distributed in both cases since it is generated from the same h_m . Hence, the simulated pre-signature is computationally indistinguishable from a real one, establishing zero-knowledge. □

Lemma 4 (Pre-verify Soundness). *If the base scheme Σ is EUF-CMA secure and HMAC-SHA256 is a secure PRF, then MWAS satisfies pre-verification soundness.*

Proof. Suppose an adversary produces $(m, Y, \hat{\sigma})$ where $\text{PreVerify}(pk, m, \hat{\sigma}) = 1$ but for which no witness w allows $\text{Adapt}(pk, m, \hat{\sigma}, w)$ to yield a valid full signature. Since PreVerify accepts, the embedded Σ -signature σ' is valid on h_m . The Adapt procedure additionally checks $\text{HMAC}_k(\langle \text{ADAPTORv1} \rangle \| \langle w \rangle) = h_{\text{com}}$ for the statement $Y = (k, h_{\text{com}})$ in $\hat{\sigma}$. If no witness satisfies this check, then h_{com} is not a valid HMAC tag for any w under key k . But k is a deterministic function of some w via SHA-256, so producing a statement Y whose h_{com} component is not an HMAC tag of any witness under the corresponding key is equivalent to forging an HMAC output. This contradicts the PRF security of HMAC [BCK96a]. Thus, the probability of breaking soundness is negligible. □

Lemma 5 (Completeness). *For every $(w, Y) \in \mathcal{R}$, honest execution of PreSign followed by Adapt yields a signature σ that is accepted by Verify except with negligible probability.*

Proof. The underlying scheme Σ ensures that a valid signature on m always verifies. Since **PreSign** correctly commits to the witness and **Adapt** incorporates the same witness into the final signature, the resulting σ is valid whenever (w, Y) is consistent. Hence, completeness holds with overwhelming probability. $\square$

Lemma 6 (Forward Security). *Exposure of a witness w linked to a statement Y does not compromise the security of past or unrelated pre-signatures.*

Proof. Revealing w allows adaptation of only those pre-signatures bound to the matching statement Y . Constructing or forging signatures for other messages requires breaking the EUF-CMA security of the base scheme Σ . Therefore, prior or unrelated pre-signatures remain unaffected, ensuring forward security. $\square$

5.4.2 Main Security Theorems

Theorem 1 (Correctness). *Let*

$$\text{MWAS} = (\text{KeyGen}, \text{GenerateStatement}, \text{PreSign}, \text{PreVerify}, \text{Adapt}, \text{Verify}, \text{Extract})$$

be our multivariate witness-hiding adaptor signature scheme. For all security parameters λ , key pairs $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$, messages $m \in \{0, 1\}^$, witnesses $w \in \mathcal{W}$ satisfying $\text{WitnessLenOK}(pk, |w|)$, and statements $Y = \text{GenerateStatement}(w)$: if $\hat{\sigma} \leftarrow \text{PreSign}(sk, m, Y)$ and $\sigma \leftarrow \text{Adapt}(pk, m, \hat{\sigma}, w)$, then*

$$\Pr [\text{Verify}(pk, m, \sigma) = 1 \wedge \text{Extract}(\hat{\sigma}, \sigma) = w] \geq 1 - \text{negl}(\lambda),$$

where the negligible failure probability originates solely from the correctness error of the base scheme Σ .

Proof. Step 1 (Pre-signature Generation). By construction, $\hat{\sigma} = (\sigma', Y, h_m)$ where:

- $m' = \langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle$, $h_m = \text{SHA}(m')$
- $\sigma' \leftarrow \Sigma.\text{Sign}(sk, h_m)$
- $Y = k \| \text{HMAC}_k(\text{"BINDING"} \| w)$, where $k = \text{SHA}(\text{"STATEMENT"} \| w)$

Correctness of Σ implies $\Pr[\Sigma.\text{Verify}(pk, h_m, \sigma') = 1] = 1$.

Step 2 (Adaptation). **Adapt** verifies the pre-signature, checks that $\text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \| \langle w \rangle) = h_{\text{com}}$ where $(k, h_{\text{com}}) = Y$, then outputs $\sigma = \langle \sigma' \rangle \| \langle w \rangle$.

Step 3 (Verification). **Verify** parses $\sigma = \langle \sigma' \rangle \| \langle w \rangle$, recomputes $Y \leftarrow \text{GenerateStatement}(w)$, recomputes $h_m = \text{SHA}(\langle m \rangle \| \langle Y \rangle \| \langle \text{PRESIGN} \rangle)$, then checks:

1. $\Sigma.\text{Verify}(pk, h_m, \sigma') = 1$ (succeeds by Step 1)

Step 4 (Extraction). **Extract** parses $\sigma = \langle \sigma'' \rangle \| \langle w \rangle$, cross-checks $\sigma'' = \sigma'$ from $\hat{\sigma}$, verifies the HMAC commitment against Y , and returns w .

The only negligible failure is from the base scheme Σ , so:

$$\Pr[\text{Verify}(pk, m, \sigma) = 1 \wedge \text{Extract}(\hat{\sigma}, \sigma) = w] \geq 1 - \text{negl}(\lambda).$$

$\square$

Theorem 2 (Witness Hiding). *For any PPT adversaries $\mathcal{A}$ possessing oracle access to $\mathcal{O}_{\text{PreSign}}(m', Y')$ for $m' \neq m$, and all witness pairs (w_0, w_1) with $|w_0| = |w_1|$, define*

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) = |\Pr[\mathcal{A}(pk, Y_0, \hat{\sigma}_0) = 1] - \Pr[\mathcal{A}(pk, Y_1, \hat{\sigma}_1) = 1]|,$$

where $Y_b = \text{GenerateStatement}(w_b)$ and $\hat{\sigma}_b = \text{PreSign}(sk, m, Y_b)$. Then,

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) \leq \text{negl}(\lambda),$$

under the PRF security of HMAC-SHA256.

Proof. We build a reduction $\mathcal{B}$ in the PRF security of HMAC. Given access to a PRF challenger that returns real HMAC outputs or random strings, $\mathcal{B}$ simulates the **PreSign** responses for $\mathcal{A}$. Distinguishing these implies breaking the PRF assumption, which is the basis of the claim. $\square$

Theorem 3 (Witness Extractability). *For any PPT adversary $\mathcal{A}$, the probability of producing a pair $(\hat{\sigma}, \sigma)$ such that*

$$\text{Verify}(pk, m, \sigma) = 1 \quad \wedge \quad \text{Extract}(\hat{\sigma}, \sigma) = \perp$$

is bounded by a negligible function in the security parameter λ :

$$\Pr[\text{Verify}(pk, m, \sigma) = 1 \quad \wedge \quad \text{Extract}(\hat{\sigma}, \sigma) = \perp] \leq \text{negl}(\lambda).$$

Proof. If extraction fails, then either:

- The HMAC tag does not match the committed witness w (breaks PRF), or
- There exist $w \neq w'$ with the same statement Y (breaks SHA-256 collision resistance)

Both contradict assumptions. Hence, extraction fails with negligible probability. $\square$

Theorem 4 (Unforgeability (aEUF-CMA)). *Let MWAS be defined over the base scheme Σ . Then for all PPT adversaries $\mathcal{A}$:*

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{aEUF-CMA}}(\lambda) \leq \text{Adv}_{\Sigma, \mathcal{B}}^{\text{EUF-CMA}}(\lambda) + \text{Adv}_{\text{HMAC}, \mathcal{C}}^{\text{PRF}}(\lambda).$$

Proof. We consider two cases:

- **Pre-signature forgery:** reduces directly to Σ via σ' .
- **Full signature forgery:** implies either forgery in Σ or breaking HMAC PRF (if pre-signature was given but modified).

Combining both, the total advantage is upper-bounded as above. $\square$

Theorem 5 (Statement Binding). *Consider any PPT adversary $\mathcal{A}$ that outputs two distinct witnesses $w \neq w'$ mapped to the same statement Y . The likelihood of such an event is upper bounded by the collision probability of the hash function used in **GenerateStatement**, namely:*

$$\Pr[(w, Y) \in \mathcal{R} \quad \wedge \quad (w', Y) \in \mathcal{R} \quad \wedge \quad w \neq w'] \leq \text{Adv}_{\text{SHA}}^{\text{CR}}(\lambda).$$

Proof. The binding of the statement implies that two distinct witnesses cannot map to the same $Y = h_1 \parallel h_2$ where $h_1 = \text{SHA}(\text{"STATEMENT"} \parallel w)$ and $h_2 = \text{HMAC}_{h_1}(\text{"BINDING"} \parallel w)$. If $w \neq w'$ but $Y = Y'$, then $h_1 = h'_1$, i.e., $\text{SHA}(\text{"STATEMENT"} \parallel w) = \text{SHA}(\text{"STATEMENT"} \parallel w')$, which is a collision in SHA-256. $\square$

5.4.3 Proof Sketches

Proof Sketch (Proof of Correctness). *The correctness follows from three points. (1) Since the underlying scheme Σ is correct by assumption, every honestly generated base signature σ' is verified under Σ with overwhelming probability. (2) The adaptation step verifies $\text{HMAC}_{\text{SHA256}}(k, \langle \text{ADAPTORv1} \rangle \| \langle w \rangle) = h_{\text{com}}$, where $(k, h_{\text{com}}) = Y$ and $(w, Y) \in \mathcal{R}$ by construction. This check always succeeds. (3) The extraction procedure parses $\sigma = \langle \sigma' \rangle \| \langle w \rangle$ and recovers w deterministically. Thus, verification and extraction fail except for a negligible probability originating from the base scheme.*

Proof Sketch (Proof of Witness Hiding). *A pre-signature $\hat{\sigma} = (\sigma', Y, h)$ contains the tag $h = \text{HMAC}_k(\text{"ADAPTORv1"} \| w)$. Since HMAC-SHA256 is assumed to be a secure $\text{PRF}[\text{BCK96b}, \text{BCK96a}]$, the distribution of h is computationally indistinguishable from uniform. Therefore, if an adversary could distinguish which witness was used, we could embed the PRF challenge into h and break the PRF assumption. Hence, witness information remains hidden to negligible advantage.*

Proof Sketch (Proof of Witness Extractability). *Given $(\hat{\sigma}, \sigma)$, where $\sigma = \langle \sigma' \rangle \| \langle w \rangle$, the witness w can be read directly by parsing σ and verified using the HMAC binding check against Y in $\hat{\sigma}$. The binding property of GenerateStatement ensures that at most one witness corresponds to a given Y . Thus, extraction always succeeds, and failure would imply that two witnesses share the same statement or two HMAC values collide — both of which are negligible under the collision resistance and PRF assumptions.*

Proof Sketch (Proof of Unforgeability). *Consider two cases. (1) Forging a pre-signature requires producing a new base signature σ' on a fresh message. This directly violates the EUF-CMA security of Σ . (2) Forging a full signature additionally requires creating a correct HMAC tag without the corresponding witness, contradicting the PRF security of $\text{HMAC}[\text{BCK96a}]$. Since both primitives are secure by assumption, the probability of forging either kind of signature is negligible.*

Proof Sketch (Proof of Statement Binding). *Suppose an adversary outputs two distinct witnesses $w \neq w'$ such that $\text{GenerateStatement}(w) = \text{GenerateStatement}(w')$. By definition, this implies the equality of their SHA-256 commitments, which constitutes a collision in the SHA-256 hash function. Since SHA-256 is assumed to be collision-resistant, the probability of such an event is negligible.*

5.4.4 Concrete Security Bounds

Theorem 6 (Concrete Security Bound). *Let $\mathcal{A}$ be any PPT adversary running in time t and making at most q oracle queries. Then the advantage of $\mathcal{A}$ in the aEUF-CMA experiment against MWAS is bounded by:*

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{aEUF-CMA}}(\lambda) \leq \text{Adv}_{\Sigma, \mathcal{B}}^{\text{EUF-CMA}}(\lambda) + \text{Adv}_{\text{HMAC}, \mathcal{C}}^{\text{PRF}}(\lambda) + \text{Adv}_{\text{SHA}, \mathcal{D}}^{\text{CR}}(\lambda),$$

where the reductions $\mathcal{B}, \mathcal{C}, \mathcal{D}$ simulate the MWAS environment with overhead at most $O(q)$ and total run time $t + O(q)$.

Proof. The bound is obtained by considering the distinct ways in which $\mathcal{A}$ could succeed:

Case 1 (Pre-signature forgery). If $\mathcal{A}$ outputs a new pre-signature on a message not queried to the oracle, then the base signature σ' inside it must be a forgery against Σ . The probability of this event is limited by $\text{Adv}_{\Sigma, \mathcal{B}}^{\text{EUF-CMA}}(\lambda)$.

Case 2 (Full signature forgery). Producing a fresh, valid signature requires either: (a) forging the embedded base signature σ' under Σ , or (b) producing a correct HMAC value without the corresponding witness, which would violate the PRF assumption on HMAC . These are captured by the first two terms of the bound.

Case 3 (Collision in statement generation). If distinct witnesses $w \neq w'$ map to the same statement Y , then a collision in SHA-256 has been found. The probability of this is given by $\text{Adv}_{\text{SHA}, \mathcal{D}}^{\text{CR}}(\lambda)$.

Bounding Advantage. Each of these cases covers a disjoint failure event. By adding the probabilities, we obtain the claimed inequality. The simulation overhead is polynomial in q and does not affect the asymptotic tightness of the results. $\square$

Theorem 7 (Individual Security Bounds). *Let $\mathcal{A}$ be any PPT adversary running in time t with at most q oracle queries. Then:*

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{WH}}(\lambda) \leq \text{Adv}_{\text{HMAC}, \mathcal{B}}^{\text{PRF}}(\lambda), \quad (5)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{EXT}}(\lambda) \leq \text{Adv}_{\text{SHA}, \mathcal{C}}^{\text{CR}}(\lambda), \quad (6)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{BIND}}(\lambda) \leq 2 \cdot \text{Adv}_{\text{SHA}, \mathcal{D}}^{\text{CR}}(\lambda), \quad (7)$$

$$\text{Adv}_{\text{MWAS}, \mathcal{A}}^{\text{FS}}(\lambda) \leq \text{Adv}_{\Sigma, \mathcal{E}}^{\text{EUF-CMA}}(\lambda). \quad (8)$$

Each reduction runs in time $t + O(q)$.

Proof. These inequalities are the result of the explicit reductions proven earlier. Witness hiding reduces to the PRF security of HMAC. Witness extractability and statement binding both reduce to the collision resistance of SHA-256; the factor of 2 in the binding bound accounts for either component of Y colliding. Forward security reduces to the unforgeability of the base scheme Σ . Each reduction maintains the polynomial overhead in time and queries. $\square$

Remark 2 (Quantum Security). The security guarantees also hold in the Quantum Random Oracle Model (QROM) [HXY19]. The base signature algorithms considered in this work are the multivariate schemes UOV and MAYO [Pat97, Beu21].

More precisely, we claim the following in the QROM:

- **Unforgeability** follows from the EUF-CMA security of the underlying MQ scheme against QPT adversaries. UOV and MAYO are designed to resist quantum attacks at their respective NIST security levels.
- **Witness Hiding** relies on the PRF security of HMAC-SHA256 against quantum distinguishers. The proof technique of embedding the PRF challenge into the commitment carries over to the QROM, since the PRF challenger and the HMAC oracle can each handle quantum superposition queries.
- **Statement Binding** relies on the collision resistance of SHA-256, which is assumed to hold against quantum adversaries (Grover’s algorithm gives at most a quadratic speedup, which is compensated by the parameter choices).

The argument that classical ROM proofs extend to the QROM when the hash function is modelled as a quantum-accessible random oracle is formalised in [BFSZ11, Zha12]. Care is required because not all ROM proof techniques transfer directly; in particular, rewinding arguments may fail. The reductions in our proofs do not rely on rewinding and are compatible with quantum access, following the framework of [Zha12].

5.5 Security Summary and Consolidation

We now consolidate the security guarantees established in the previous theorems and provide a comprehensive summary of the security properties of the MWAS construction.

Security Assumptions. Our MWAS construction relies on the following cryptographic assumptions.

1. The base multivariate signature scheme $\Sigma \in \{\text{UOV}, \text{MAYO}\}$ is existentially unforgeable under chosen-message attacks (EUF-CMA) in the ROM.
2. HMAC-SHA256 is a secure pseudorandom function (PRF) against polynomial-time adversaries.
3. SHA-256 exhibits collision resistance against polynomial-time adversaries.

Security Characteristics. Based on our assumptions, the MWAS framework guarantees the following properties:

- **Validity:** Whenever a pre-signature $\hat{\sigma}$ is produced honestly together with its witness w , the adaptation procedure generates a signature σ that is accepted by the verification algorithm except with negligible error. Moreover, the extraction routine deterministically produces the same witness w .
- **Hidden Witness:** Pre-signatures constructed from different witnesses cannot be efficiently distinguished by any adversary, even if the adversary has access to the pre-signing oracle. This indistinguishability reduces to the pseudo-randomness of HMAC-SHA256 [BCK96a].
- **Recoverability:** From any valid pre-signature and its completed signature, there exists an efficient procedure that identifies the corresponding witness. The chance of failure in recovering the correct witness is negligible.
- **Single-Witness Consistency:** Each valid pre-signature is linked to at most one witness that can produce a valid adapted signature, ruling out the possibility of multiple different witnesses being compatible.
- **Resistance to Forgery (aEUF-CMA):** No efficient adversary can generate a fresh pre-signature or full signature on a message outside its queries. Any such attempt would either compromise the security of the underlying EUF-CMA scheme or contradict the pseudorandomness of HMAC.
- **Statement Soundness:** It is infeasible to construct two different witnesses that correspond to the same public statement. Any adversary capable of this would necessarily violate the assumed collision resistance of SHA-256.
- **Forward Protection:** Learning a witness w for one specific pre-signature does not compromise other pre-signatures created in the past or for unrelated messages. Security in this sense follows from the EUF-CMA guarantees of the base scheme.

Consolidated Security Theorem.

Theorem 8 (Comprehensive Security Guarantee). *Assume the underlying multivariate signature scheme [Pat97, Beu21, DY09] is secure against existential forgery under chosen-message attacks (EUF-CMA) in the ROM. Further, assume that HMAC-SHA256 behaves as a secure PRF [BCK96b, BCK96a] and that SHA-256 resists collisions. Under these conditions, MWAS satisfies Correctness (Def. 1), Witness Hiding (Def. 4), Witness Recovery (Def. 5), Witness Uniqueness (Def. 6), Statement Binding (Def. 7), Pre-signature Privacy (Def. 8), Adaptation Soundness (Def. 9), and adaptive unforgeability (aEUF-CMA, Def. 10). These guarantees hold in the ROM; they extend to the QROM under the same assumptions (see Remark 2).*

6 Parameter Study and Empirical Results

6.1 Parameter Values and Signature Lengths

In the MWAS framework, the overall signature length is determined by contributions from both the chosen multivariate base scheme and the adaptor components. Table 2 reports, for each parameter set, the size of the pre-signature $\hat{\sigma}$ (equal to the raw base-scheme signature σ') and the completed signature $|\sigma| = |\sigma'| + |w|$, where w is the scheme-specific witness (see Table 1). All sizes are in bytes.

Table 2: MWAS signature sizes across all parameter sets (bytes). Pre-sig sizes report the raw base-scheme signature σ' only; the 64-byte statement Y is established in a one-time setup exchange and is not retransmitted per pre-signature.

Scheme	Pre-sig $ \hat{\sigma} $	Signature $ \sigma $	NIST Level
UOV-128	96	144	I
UOV-192	128	192	III
UOV-256	200	280	V
MAYO-128	454	478	I
MAYO-192	681	713	III
MAYO-256	964	1004	V

6.2 Comparison with Existing Schemes

Table 3 contrasts our constructions with prior post-quantum adaptor signatures. Lattice-based [EEE20], isogeny-based [TMSM21b], and code-based [KH22] schemes exhibit pre-signatures up to 19 KB and signatures in the kilobyte range. In comparison, our UOV- and MAYO-based schemes reduce sizes by one to two orders of magnitude, with UOV-128 achieving a 96-byte pre-signature and 144-byte final signature.

Table 3: Comparison of adaptor signature sizes (bytes).

Scheme	Pre-Signature	Signature
[TMSM21b]	$18327 \leq \hat{\sigma} \leq 19944$	$263 \leq \sigma \leq 1880$
[EEE20]	3210	3210
[KH22]	2098	2098
Our UOV-128	96	144
Our UOV-192	128	192
Our UOV-256	200	280
Our MAYO-128	454	478
Our MAYO-192	681	713
Our MAYO-256	964	1004

Size Improvements. MWAS achieves significant size reductions compared to prior works:

- **UOV-128:** $22\times$ smaller pre-signatures, $15\times$ smaller signatures vs. [KH22].
- **UOV-192:** $16\times$ smaller pre-signatures, $11\times$ smaller signatures vs. [KH22].
- **UOV-256:** $10\times$ smaller pre-signatures, $7\times$ smaller signatures vs. [KH22].
- **MAYO-128:** $5\times$ smaller pre-signatures, $4\times$ smaller signatures vs. [KH22].
- **MAYO-192:** $3\times$ smaller pre-signatures, $3\times$ smaller signatures vs. [KH22].

- **MAYO-256:** $2\times$ smaller pre-signatures, $2\times$ smaller signatures vs. [KH22].

Overall, MWAS achieves substantially smaller pre-signatures than prior post-quantum adaptor signatures while keeping final signature sizes competitive, which is favourable for bandwidth- and storage-constrained environments.

6.3 Software Implementation and Benchmarks

Implementation. MWAS was implemented in C using `liboqs 0.14.1-dev` (UOV/MAYO) and OpenSSL 3.0.17 for HMAC-SHA256. All operations are constant-time with secure memory handling.

Platform. The experiments were carried out on a Raspberry Pi 5 equipped with a Broadcom BCM2712 2.4 GHz quad-core Cortex-A76 CPU, 16 GB RAM, and Linux kernel 6.12.47. All code was compiled with GCC 12.2.0 using the flags `-O3 -march=native -DNDEBUG`. This setup represents a resource-constrained embedded device, providing a conservative performance baseline.

Code Availability. The complete implementation, including source code and experimental scripts, is available at: <https://github.com/Ash-OblivionKey/multivariate-adaptor-signatures.git>

Performance Analysis. Table 4 compares computational timings across MWAS and prior post-quantum adaptor signature schemes. All measurements are the mean of 100 end-to-end iterations, preceded by 10 warm-up runs. The underlying UOV and MAYO operations use the `liboqs` library [SM16, AASA⁺20].

Table 4: Measured runtimes (ms) of MWAS components across UOV, MAYO, and reference schemes.

Scheme	Parameters	KeyGen	PreSig	PreVer	Sign	Verify	Adapt	Extract
[TMSM21b]	$(2^1, 56, 16)$	151	14,709	14,780	3105	3032	5.07	5.09
	$(2^2, 38, 14)$	268	13,747	13,612	2077	2003	4.86	4.95
	$(2^3, 28, 16)$	473	12,747	12,636	1552	1491	4.74	4.91
	$(2^4, 23, 13)$	929	13,247	13,217	1290	1238	4.76	4.87
	$(2^5, 16, 16)$	3411	12,267	12,164	925	869	4.88	4.77
	$(2^8, 13, 11)$	13,170	12,448	12,310	723	679	4.73	4.64
	$(2^{10}, 11, 7)$	56,174	13,862	13,512	717	607	5.13	5.07
[KH22]	McEliece	2553	2098	349	2098	349	0.13	0.11
UOV-128	OV-Is	3.3	0.13	0.03	0.09	0.03	0.003	0.002
UOV-192	OV-Ip	1.7	0.09	0.03	0.06	0.04	0.003	0.002
UOV-256	OV-III	10.5	0.31	0.12	0.26	0.12	0.003	0.002
MAYO-128	M-1	0.14	0.55	0.17	0.38	0.17	0.003	0.003
MAYO-192	M-3	0.38	1.46	0.46	1.00	0.46	0.003	0.003
MAYO-256	M-5	0.90	3.21	0.98	2.23	0.98	0.003	0.003

Performance Insights. Adaptation is constant-time at 0.003 ms across all security levels, as it consists solely of an HMAC check and concatenation. Pre-signature generation scales with the base signing cost: UOV-128 takes 0.13 ms versus 0.31 ms for UOV-256, while MAYO ranges from 0.55 ms (MAYO-128) to 3.21 ms (MAYO-256). Compared to prior post-quantum adaptor signatures [TMSM21b, EEE20, KH22], MWAS pre-signature generation is two to five orders of magnitude faster and pre-signature sizes are $2\text{--}22\times$ smaller, primarily because the adaptor overhead (commitment + witness) is decoupled from the base signature size.

6.4 Performance Under Network Latency

To assess the practicality of our MWAS construction in realistic conditions, we evaluated its performance under varying network delays of 30 ms, 120 ms, 225 ms, and 320 ms. Four key metrics were measured across all parameter sets (UOV-128/192/256, MAYO-128/192/256): (i) end-to-end execution time, (ii) throughput, (iii) peak memory usage, and (iv) stability score. Each experiment was conducted with 1000 iterations per latency condition, preceded by 10 warmup runs to ensure consistent performance measurements.

End-to-End Time. Execution times remained essentially constant across all network latency conditions, demonstrating that MWAS operations are primarily CPU-bound rather than network-bound. MAYO-128 achieved the lowest end-to-end execution time at 1.41 ms, followed by UOV-192 at 1.86 ms and UOV-128 at 3.55 ms. The heavier variants (UOV-256, MAYO-256) extended up to 11.41 ms and 8.31 ms respectively. This shows that MWAS maintains low response times across all security levels, with network latency having negligible impact on total execution time (see Fig. 2).

Throughput. The throughput measurements remained remarkably stable under all latency conditions. MAYO-128 achieved the highest throughput at approximately 710 ops/s, followed by UOV-192 at 537 ops/s. UOV-128 achieved 282 ops/s, while MAYO-192 and MAYO-256 achieved 267 ops/s and 120 ops/s respectively. The 256-bit UOV scheme demonstrated the lowest throughput at 88 ops/s. All schemes maintained consistent throughput regardless of network latency, confirming that MWAS performance is not affected by network conditions within the tested range (Fig. 1).

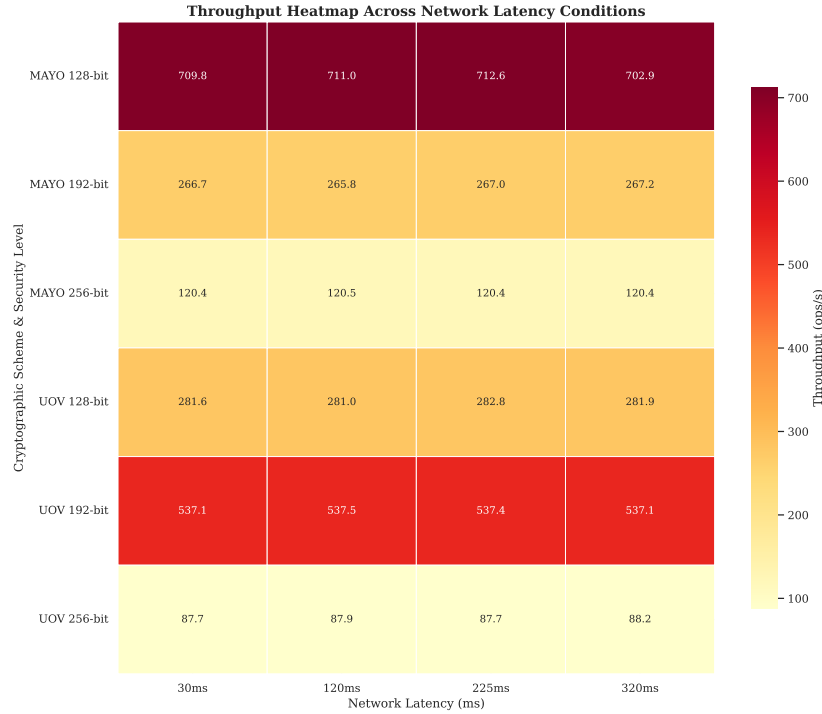


Figure 1: Throughput heatmap of MWAS schemes across varying network latencies.

Memory Usage. The reported memory figures correspond to the process Resident Set Size (RSS) measured at peak during the full benchmark run. RSS includes: shared libraries

loaded at process startup (liboqs, OpenSSL), large public-key structures held in memory across iterations, and runtime buffers used during signing and verification. Because RSS grows with security level and scheme-specific data structures, values naturally scale with the parameter set. The *per-operation* memory footprint is significantly smaller than the peak RSS figure. These values should therefore be interpreted as the total process working set, not the incremental cost of a single adaptor signature operation.

Peak memory consumption remained constant for all network latency conditions. UOV schemes consistently required less memory than their MAYO counterparts at equivalent security levels. UOV-128 used 537 MB, UOV-192 used 1,766 MB, and UOV-256 used 4,405 MB. MAYO schemes required significantly more memory: MAYO-128 used 2,733 MB, MAYO-192 used 6,410 MB, and MAYO-256 required the most at 13,964 MB. The memory footprint scaled predictably with the security level, simplifying deployment planning.

Stability. The *stability score* is defined as $100 \times (1 - CV)$, where $CV = \sigma_t / \mu_t$ is the coefficient of variation (ratio of standard deviation to mean execution time). A score of 100 indicates perfectly consistent timing; scores above 95 indicate low variance.

All MWAS variants achieved high stability scores across latency conditions, with most schemes maintaining stability above 99%. UOV-192 showed the lowest stability at 94.9%, followed by UOV-128 at 95.3%. MAYO schemes demonstrated excellent stability: MAYO-128 at 99.7%, MAYO-192 at 99.6%, and MAYO-256 at 99.7%. UOV-256 demonstrated the highest stability at 99.4%, remaining effectively stable across all tested conditions.

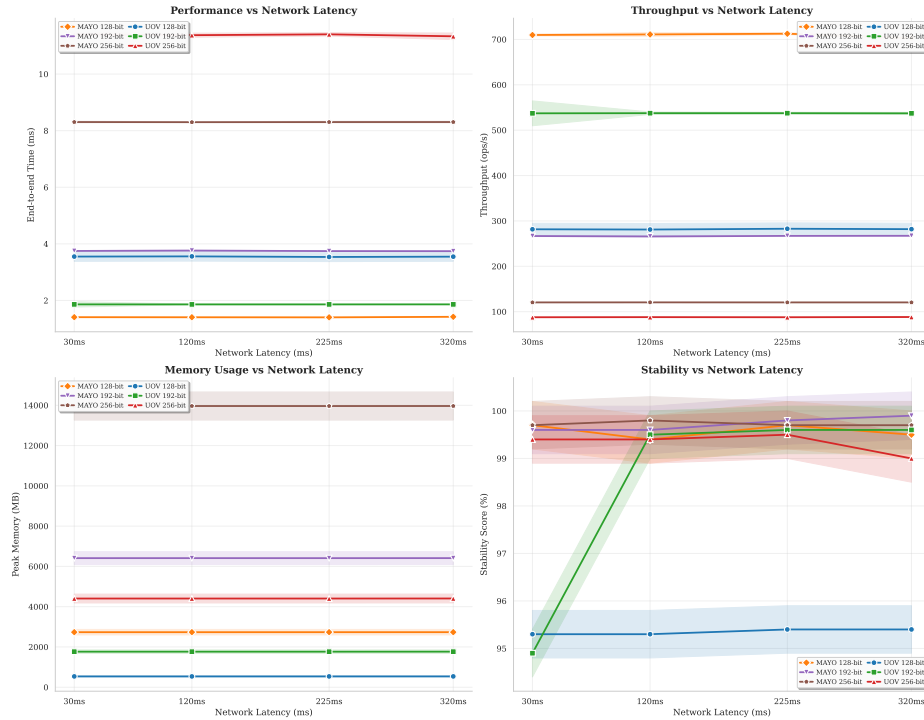


Figure 2: Comprehensive MWAS performance across network latencies: End-to-end time, throughput, memory usage, and stability.

In summary, MWAS performance is CPU-bound and insensitive to network latency within the tested range. UOV-192 offers the best balance of speed and security (537 ops/s, 1.86 ms end-to-end), while MAYO-128 provides the highest throughput (710 ops/s) at Level I security.

7 Discussion

The Multivariate Witness-Hiding Adaptor Signature (MWAS) framework advances post-quantum adaptor signatures by combining compactness, efficiency, and strong security guarantees. Our construction reduces pre-signature sizes to as little as 96 bytes (UOV-128), achieving reductions of two orders of magnitude over prior proposals and directly addressing the communication overhead that has long hindered adaptor signatures in bandwidth- and storage-constrained environments.

Scope of Applications. MWAS supports a *hash-preimage relation*: the statement Y is derived from the witness w via HMAC-SHA256, and extraction recovers w from the concatenated signature. This design is well-suited to applications where the secret being exchanged is a hash preimage, such as:

- **Fair exchange of secrets:** two parties exchange secret values conditioned on signature validity.
- **Shared-secret protocols:** a pre-signature commits to a shared secret; completion releases it.
- **Off-chain and layer-2 settings:** where signatures are exchanged between trusted parties without on-chain size constraints.

MWAS does *not* directly support discrete-logarithm or lattice-based relations used in Bitcoin Schnorr adaptor signatures or hybrid lattice protocols. Furthermore, UOV public keys (up to 1.2 MB) exceed the legacy Bitcoin block size limit (≈ 1 MB base block size), making on-chain UOV deployment infeasible without protocol-level changes. MAYO public keys (1.4 KB) are substantially more compact and are better suited to layer-2 or off-chain settings. We therefore position MWAS as a post-quantum primitive for environments where hash-preimage witnesses are natural and where on-chain storage is not required.

Performance results are reported in Section 6. The key finding relevant to deployment is that MWAS operations are CPU-bound and insensitive to network latency, which means performance on local hardware directly predicts performance in networked settings.

Nevertheless, MWAS inherits inevitable trade-offs. Public keys remain large, ranging from tens of kilobytes to more than a megabyte, which complicates their distribution in key-intensive protocols. Unlike lattice-based schemes [TSM21b, EEE20], multivariate schemes, although widely studied, lack equally tight reductions and comparably conservative security margins. Additionally, proofs in the ROM entail nontrivial loss factors. Signature sizes, though drastically reduced compared to existing post-quantum adaptor signatures, exceed those of traditional methods like ECDSA [JMV01], which must be weighed in deployment contexts.

These limitations open promising directions for future work. Structured key compression may substantially reduce public key sizes. Threshold and batch-verifiable variants could extend utility to high-throughput and distributed settings. Hardware acceleration and embedded optimisations would expand the applicability in resource-constrained environments. On the theoretical side, developing tighter reductions and exploring standard model guarantees would strengthen long-term confidence in multivariate-based constructions.

MWAS demonstrates that adaptor signatures can be instantiated from MQ problems without significant overhead, and that the commit-then-sign paradigm is a viable substitute for algebraic witness embedding when the latter is unavailable. This broadens the design space of post-quantum adaptor signatures beyond lattice-based assumptions.

8 Conclusion

This work introduces the first adaptor-style signature scheme instantiated over MQ signatures, establishing MWAS as both a practical and theoretically grounded alternative to lattice-based post-quantum designs. Building on the mature cryptanalytic foundations of UOV and MAYO, we show that multivariate signatures can support adaptor functionality with competitive efficiency and dramatically reduced communication overhead. Our implementation achieves pre-signatures ranging from 96 bytes (UOV-128) to 964 bytes (MAYO-256) and generation times under 0.32 ms for UOV variants. In comparison, MAYO-based instantiations balance moderately larger signatures (454–964 bytes) with significantly reduced key sizes. Adaptation and extraction operations incur negligible cost (0.002–0.003 ms), ensuring minimal overhead relative to the base signature schemes. Validation across a comprehensive test suite confirms correctness, stability, and constant-time execution, reinforcing side-channel resistance. Importantly, performance is sustained on resource-constrained hardware and remains consistent under adverse network conditions. Beyond concrete performance, MWAS contributes to cryptographic diversity. The dominance of lattice-based adaptor signatures concentrates systemic risk: breakthroughs against lattice assumptions would threaten an entire class of deployed systems. By establishing MQ problems as a viable foundation for adaptor signatures, we broaden the ecosystem of post-quantum primitives and mitigate reliance on a single hardness assumption. Benchmarks show pre-signature generation over $114,000\times$ faster than [TMSM21b] (0.13 ms vs. 14,708 ms) and pre-signature sizes up to $22\times$ smaller than [KH22] (96 bytes vs. 2,098 bytes), making MWAS well-suited for bandwidth- and latency-sensitive deployments.

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